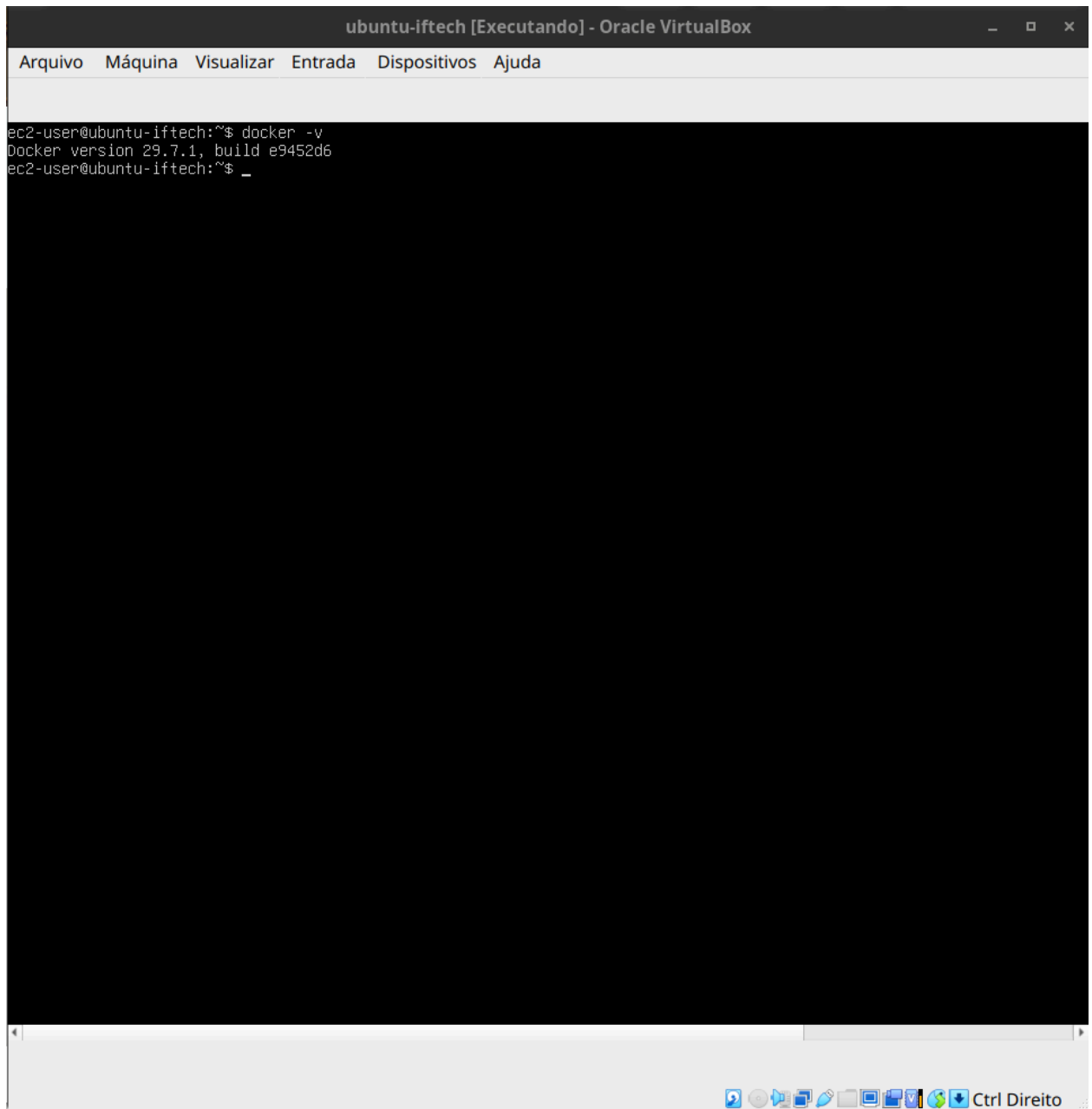


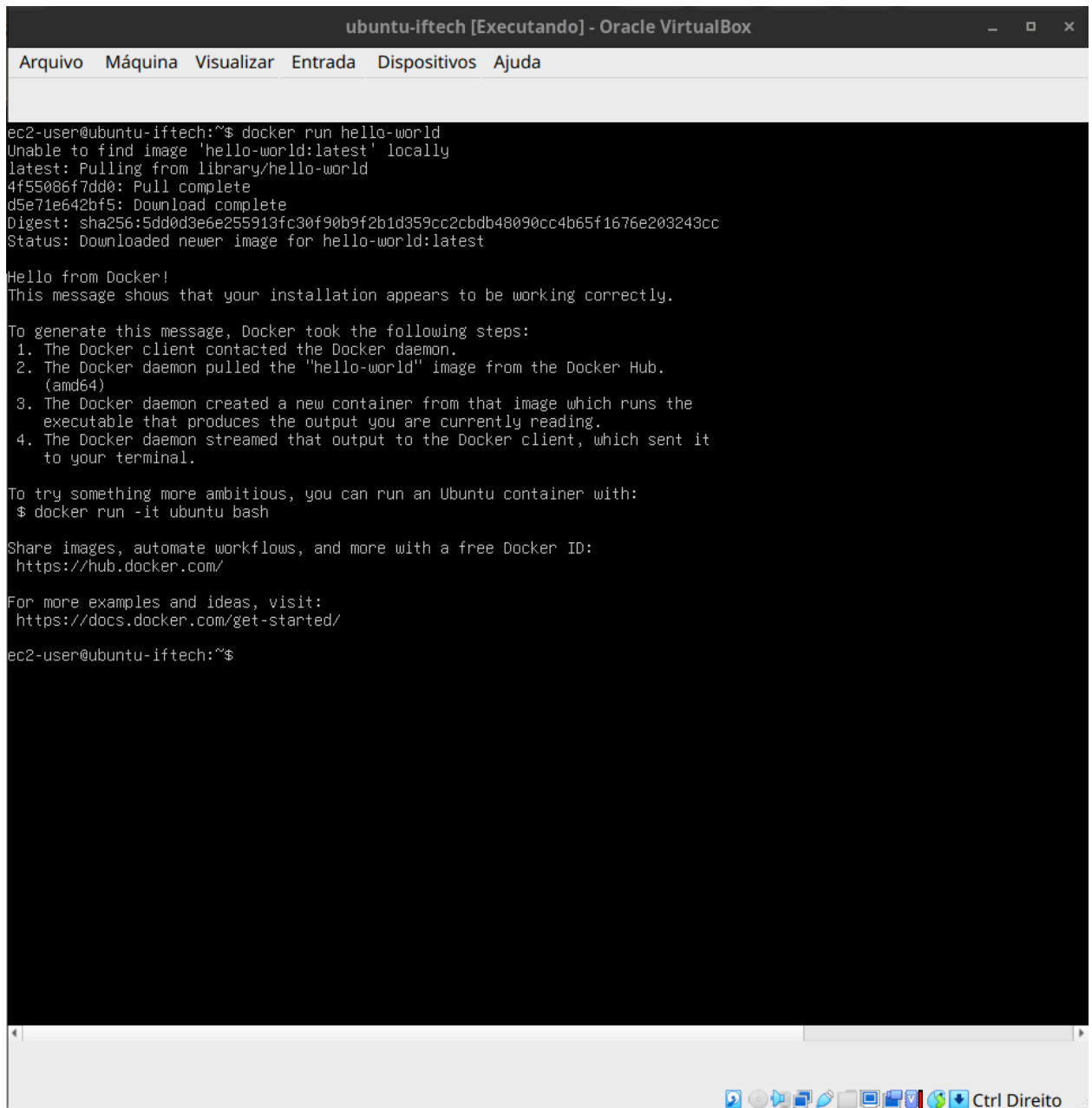
1) Instalar o docker



The screenshot shows a terminal window titled "ubuntu-iftech [Executando] - Oracle VirtualBox". The window has a menu bar with "Arquivo", "Máquina", "Visualizar", "Entrada", "Dispositivos", and "Ajuda". The terminal output shows the command "docker -v" being executed, resulting in the output "Docker version 29.7.1, build e9452d6". The prompt "ec2-user@ubuntu-iftech:~\$" is visible at the end of the line.

```
ec2-user@ubuntu-iftech:~$ docker -v
Docker version 29.7.1, build e9452d6
ec2-user@ubuntu-iftech:~$ _
```

2) Baixar e executar container



```
ubuntu-iftech [Executando] - Oracle VirtualBox
Arquivo  Máquina  Visualizar  Entrada  Dispositivos  Ajuda

ec2-user@ubuntu-iftech:~$ docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:5dd0d3e6e255913fc30f90b9f2b1d359cc2cbdb48090cc4b65f1676e203243cc
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

ec2-user@ubuntu-iftech:~$
```

3) Baixar e executar container sem prender o terminal com mapeamento de porta

```
ubuntu-iftech [Executando] - Oracle VirtualBox
Arquivo  Máquina  Visualizar  Entrada  Dispositivos  Ajuda

latest: Pulling from dockersamples/static-site
aff3ab7e9c39: Pull complete
fdd5d7827f33: Pull complete
a3ed95caeb02: Pull complete
716f7a5f3082: Pull complete
7b10f03a0309: Pull complete
Digest: sha256:daa686c61d7d239b7977e72157997489db49f316b9b9af3909d9f10fd28b2dec
Status: Downloaded newer image for dockersamples/static-site:latest
19deb58d6c8d0987c72367a90b32430c533491e2546d69b14372167c4cb54508
ec2-user@ubuntu-iftech:~$ docker ps
CONTAINER ID   IMAGE                COMMAND                  CREATED        STATUS        PORTS
19deb58d6c8d   dockersamples/static-site  "/bin/sh -c 'cd /usr..."  5 seconds ago  Up 4 seconds  443/tcp, 0.0.0.0:8080->8080
arahamihira
ec2-user@ubuntu-iftech:~$ wget http://localhost:8080/
--2026-08-18 11:08:14--  http://localhost:8080/
Resolving localhost (localhost)... ::1, 127.0.0.1
Connecting to localhost (localhost)|::1|:8080... connected.
HTTP request sent, awaiting response... 200 OK
Length: 8749 (8,5K) [text/html]
Saving to: 'index.html'

index.html                                     100%[=====]
2026-08-18 11:08:14 (429 MB/s) - 'index.html' saved [8749/8749]

ec2-user@ubuntu-iftech:~$ ls
index.html
ec2-user@ubuntu-iftech:~$ head -n 5 index.html
<!DOCTYPE html><html>

<head>
<meta charset="utf-8">
<style type="text/css">
ec2-user@ubuntu-iftech:~$ head -n 15 index.html
<!DOCTYPE html><html>

<head>
<meta charset="utf-8">
<style type="text/css">
body {
  font-family: Helvetica, arial, sans-serif;
  font-size: 14px;
  line-height: 1.6;
  padding-top: 10px;
  padding-bottom: 10px;
  background-color: white;
  padding: 30px; }

body > *:first-child {
ec2-user@ubuntu-iftech:~$
```

4) Baixando a imagem do ubuntu

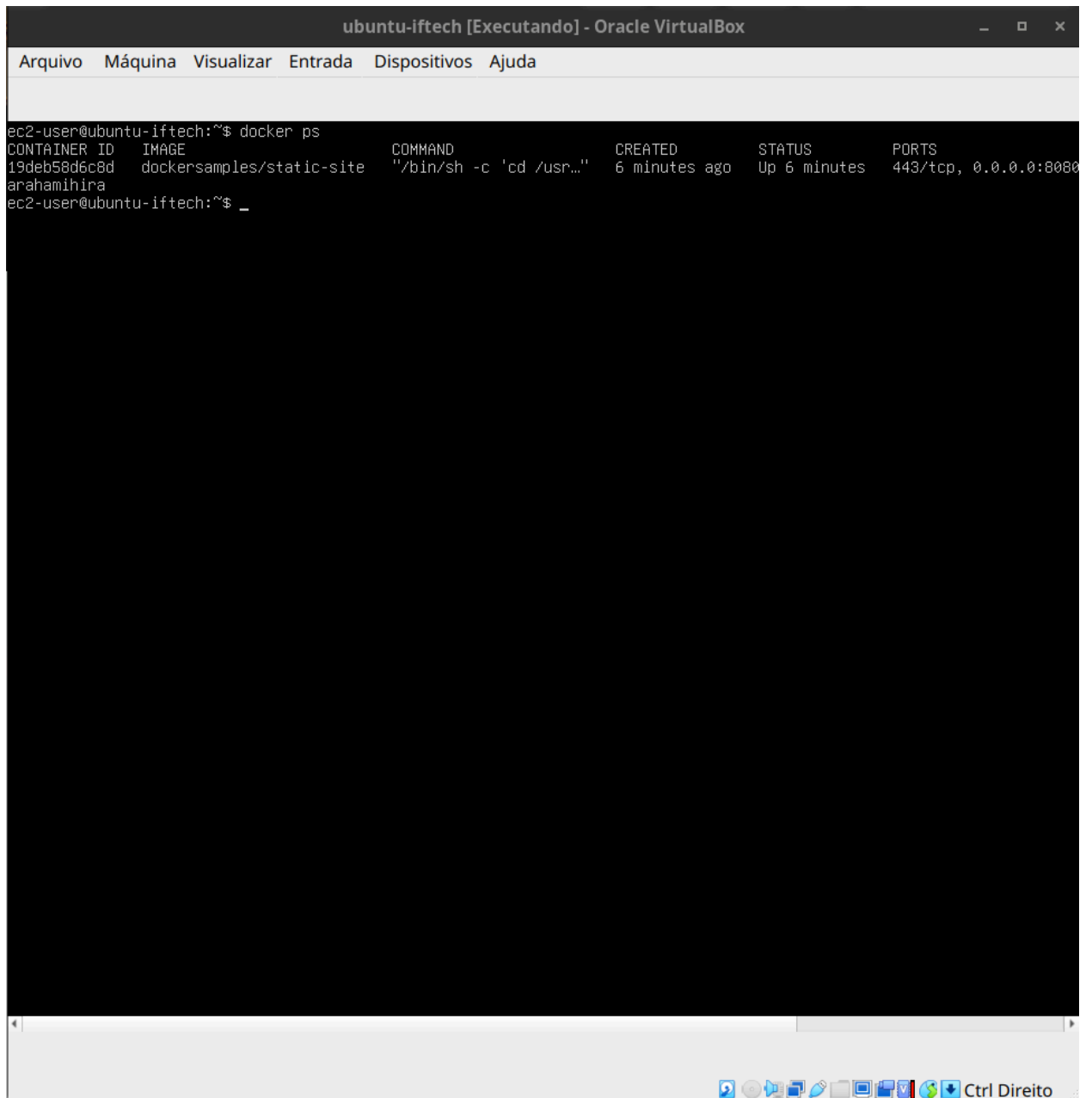
```
ubuntu-iftech [Executando] - Oracle VirtualBox
Arquivo  Máquina  Visualizar  Entrada  Dispositivos  Ajuda

ec2-user@ubuntu-iftech:~$ docker pull ubuntu:latest
latest: Pulling from library/ubuntu
a7fb98a8eddd: Pull complete
617772c7d19b: Pull complete
failed to do request: Get "https://registry-1.docker.io/v2/library/ubuntu/referers/sha256:522d96ea380735b5cf03988f84c2a...al tcp [2600:1f18:2148:bc01:6bf8:dabe:969c:e9c4]:443: connect: network is unreachable
ec2-user@ubuntu-iftech:~$ docker image ls

IMAGE                                ID                                DISK USAGE  CONTENT SIZE  EXTRA
app-backend:latest                  f8ff70b66042                      1.46GB      312MB
app-frontend:latest                 621c126a5431                      794MB       182MB
dockersamples/static-site:latest    daa686c61d7d                      293MB       74.9MB      U
hello-world:latest                  5dd0d3e6e255                      25.9kB      9.49kB      U
ollama/ollama:latest                4dea9fb51194                      8.04GB      3.26GB

ec2-user@ubuntu-iftech:~$ docker pull ubuntu:latest
latest: Pulling from library/ubuntu
```

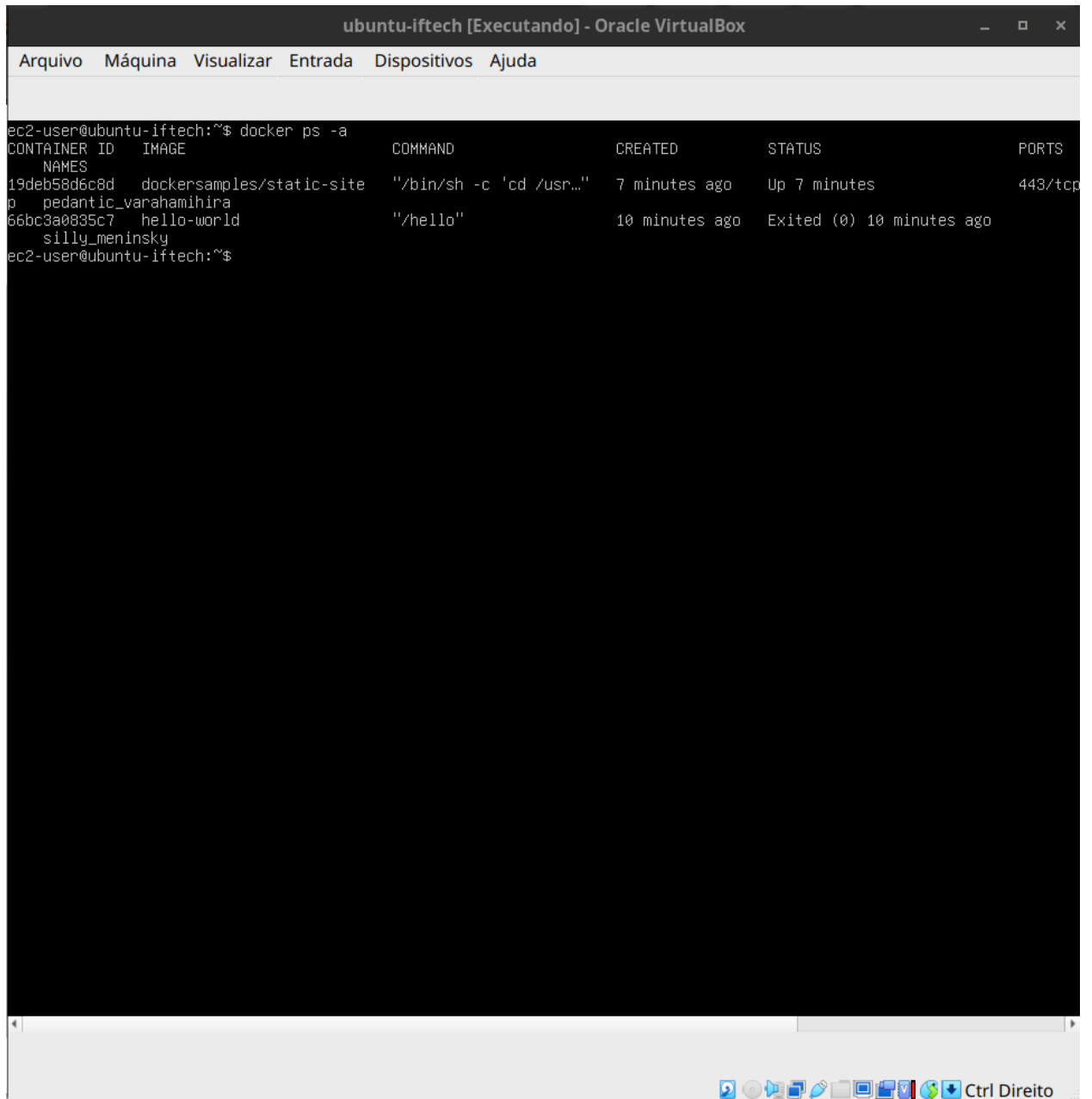
- 5) Verificar os containers em execução. Veja que apenas a imagem foi baixada.



```
ubuntu-iftech [Executando] - Oracle VirtualBox
Arquivo  Máquina  Visualizar  Entrada  Dispositivos  Ajuda

ec2-user@ubuntu-iftech:~$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
19deb58d6c8d   dockersamples/static-site          "/bin/sh -c 'cd /usr..." 6 minutes ago  Up 6 minutes  443/tcp, 0.0.0.0:8080
arahamihira
ec2-user@ubuntu-iftech:~$ _
```

- 6) Verificar os container em execução e os que não estão mais em execução. Veja que o container hello-world não estava em execução anteriormente e está listado.

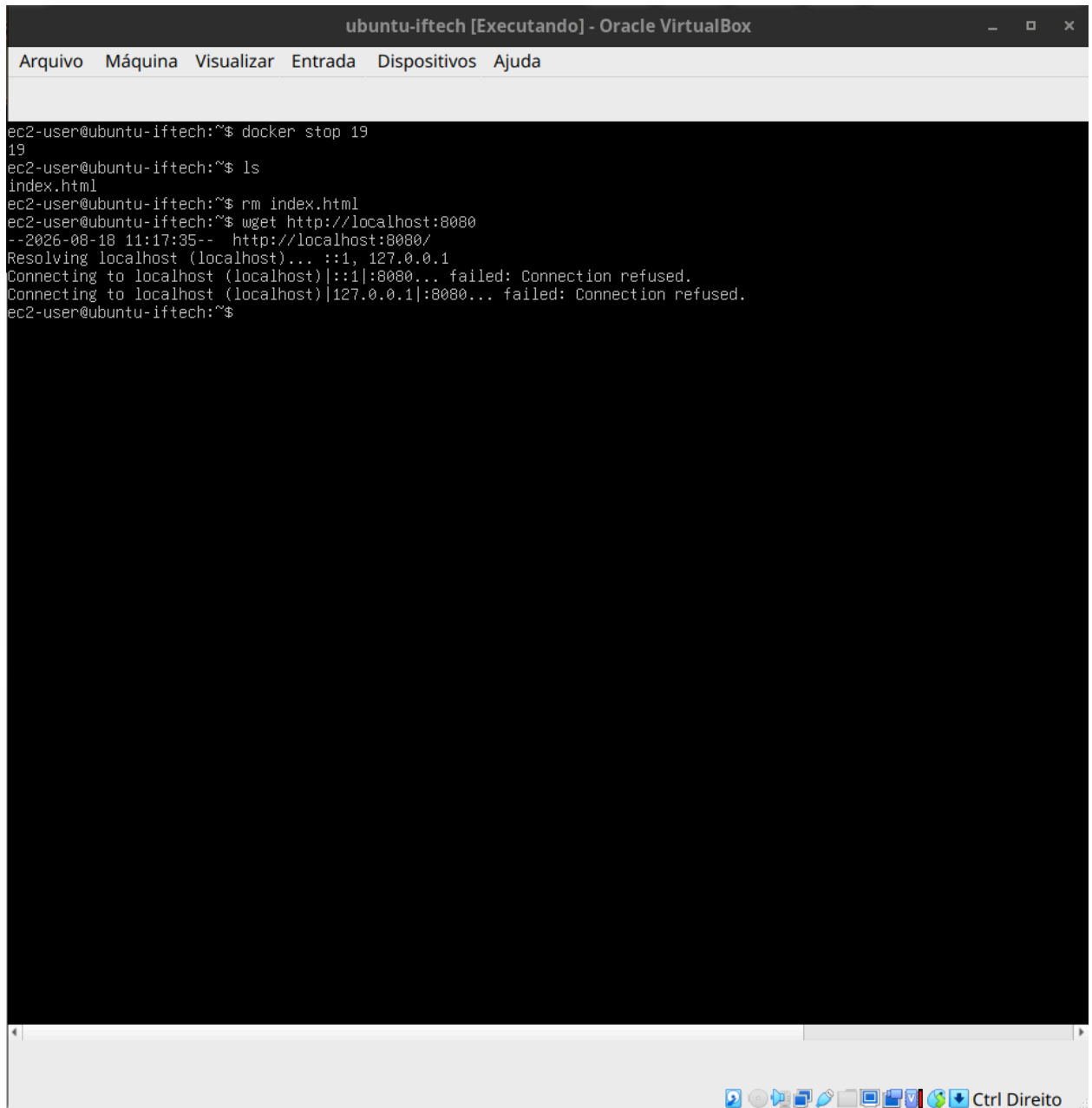


```
ec2-user@ubuntu-iftech:~$ docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
19deb58d6c8d	dockersamples/static-site	"/bin/sh -c 'cd /usr...'"	7 minutes ago	Up 7 minutes	443/tcp
p_pedantic_varahamihira	dockersamples/static-site	"/bin/sh -c 'cd /usr...'"	7 minutes ago	Up 7 minutes	443/tcp
66bc3a0835c7	hello-world	"/hello"	10 minutes ago	Exited (0) 10 minutes ago	

```
ec2-user@ubuntu-iftech:~$
```

- 7) Parar um container a partir de um ID, teste com o ID do container dockersamples/static-site do item 3.



```
ubuntu-iftech [Executando] - Oracle VirtualBox
Arquivo  Máquina  Visualizar  Entrada  Dispositivos  Ajuda

ec2-user@ubuntu-iftech:~$ docker stop 19
19
ec2-user@ubuntu-iftech:~$ ls
index.html
ec2-user@ubuntu-iftech:~$ rm index.html
ec2-user@ubuntu-iftech:~$ wget http://localhost:8080/
--2026-08-18 11:17:35--  http://localhost:8080/
Resolving localhost (localhost)... ::1, 127.0.0.1
Connecting to localhost (localhost)|::1|:8080... failed: Connection refused.
Connecting to localhost (localhost)|127.0.0.1|:8080... failed: Connection refused.
ec2-user@ubuntu-iftech:~$
```

- 8) Inicie o container a partir de um ID, teste com o ID do container dockersamples/static-site do item 7 que foi parado

```
ubuntu-iftech [Executando] - Oracle VirtualBox
Arquivo  Máquina  Visualizar  Entrada  Dispositivos  Ajuda

ec2-user@ubuntu-iftech:~$ docker ps -a
CONTAINER ID   IMAGE                COMMAND                  CREATED        STATUS
19deb58d6c8d   dockersamples/static-site  "/bin/sh -c 'cd /usr..."  10 minutes ago  Exited (137) About a minute ago
66bc3a0835c7   hello-world          "/hello"                 12 minutes ago  Exited (0) 12 minutes ago
ec2-user@ubuntu-iftech:~$ docker start 19
19
ec2-user@ubuntu-iftech:~$ wget http://localhost:8080
--2026-08-18 11:18:22--  http://localhost:8080/
Resolving localhost (localhost)... ::1, 127.0.0.1
Connecting to localhost (localhost)|::1|:8080... connected.
HTTP request sent, awaiting response... 200 OK
Length: 8749 (8,5K) [text/html]
Saving to: 'index.html'

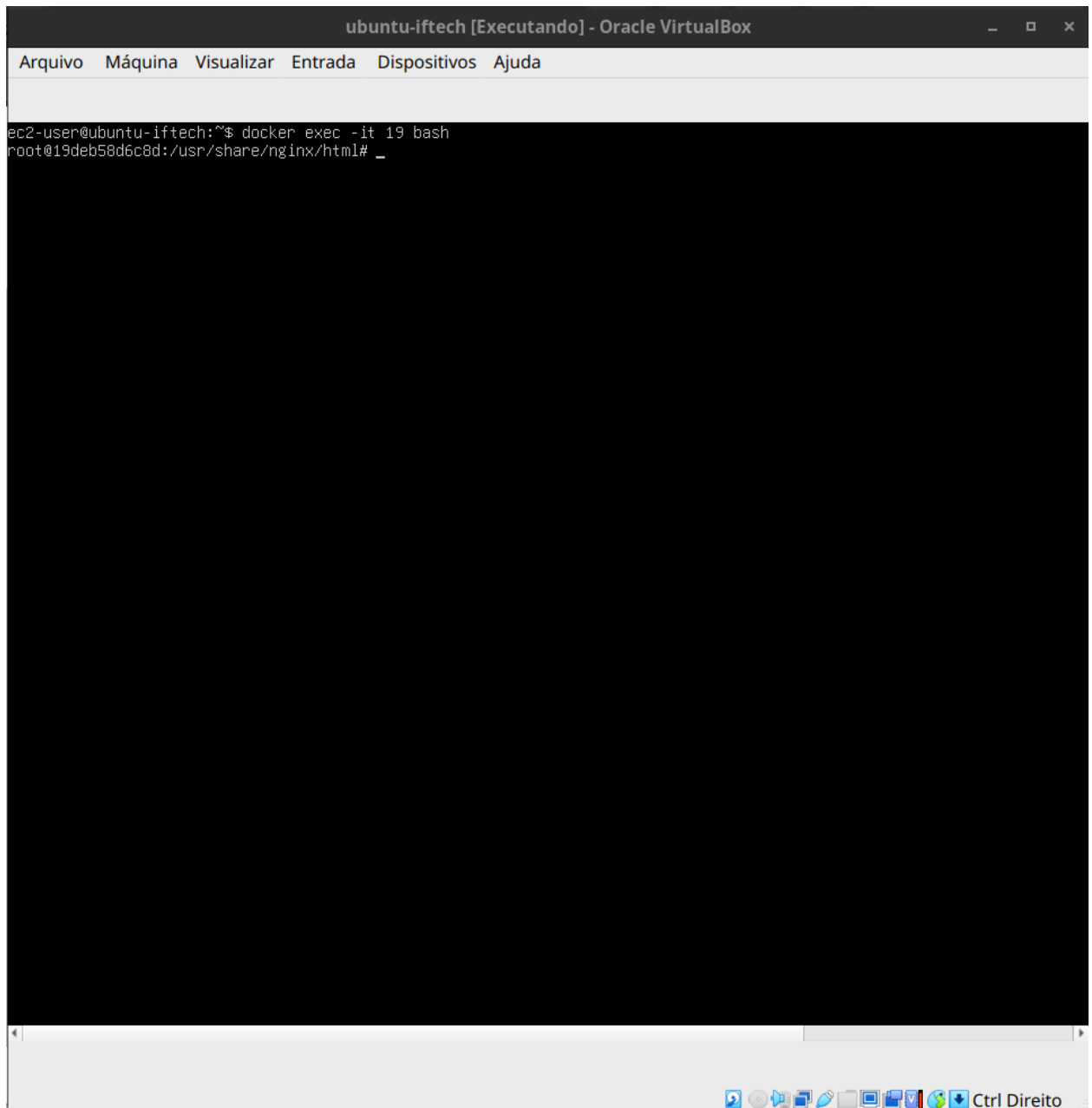
index.html                                     100%[=====]

2026-08-18 11:18:22 (60,3 MB/s) - 'index.html' saved [8749/8749]

ec2-user@ubuntu-iftech:~$ head -n 10 index.html
<!DOCTYPE html><html>

<head>
<meta charset="utf-8">
<style type="text/css">
body {
  font-family: Helvetica, arial, sans-serif;
  font-size: 14px;
  line-height: 1.6;
  padding-top: 10px;
ec2-user@ubuntu-iftech:~$ _
```

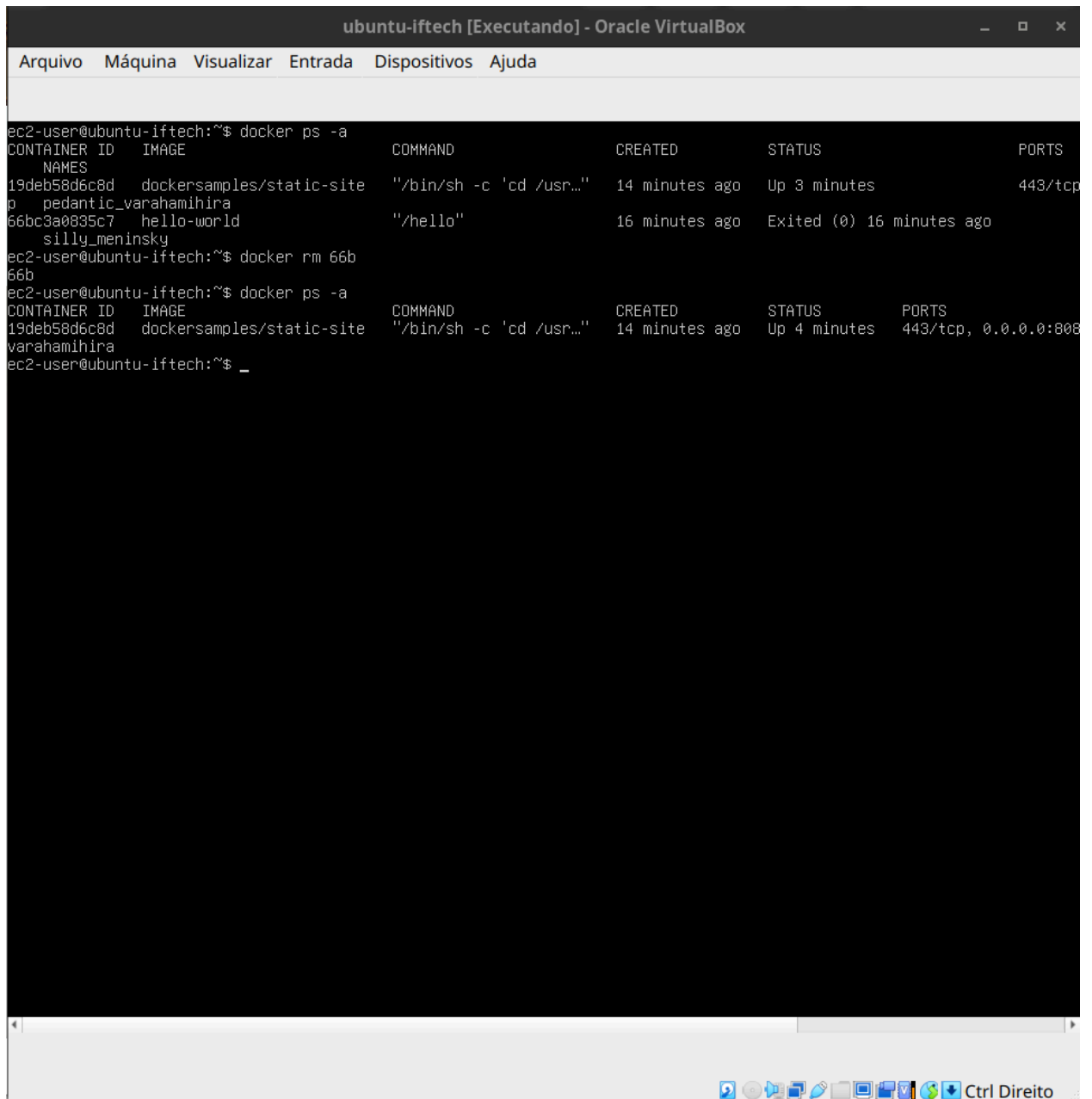

- 9) Executar e entrar em um bash de um container com ID dele, use o id do item 8



```
ubuntu-iftech [Executando] - Oracle VirtualBox
Arquivo  Máquina  Visualizar  Entrada  Dispositivos  Ajuda

ec2-user@ubuntu-iftech:~$ docker exec -it 19 bash
root@19deb58d6c8d:/usr/share/nginx/html# _
```

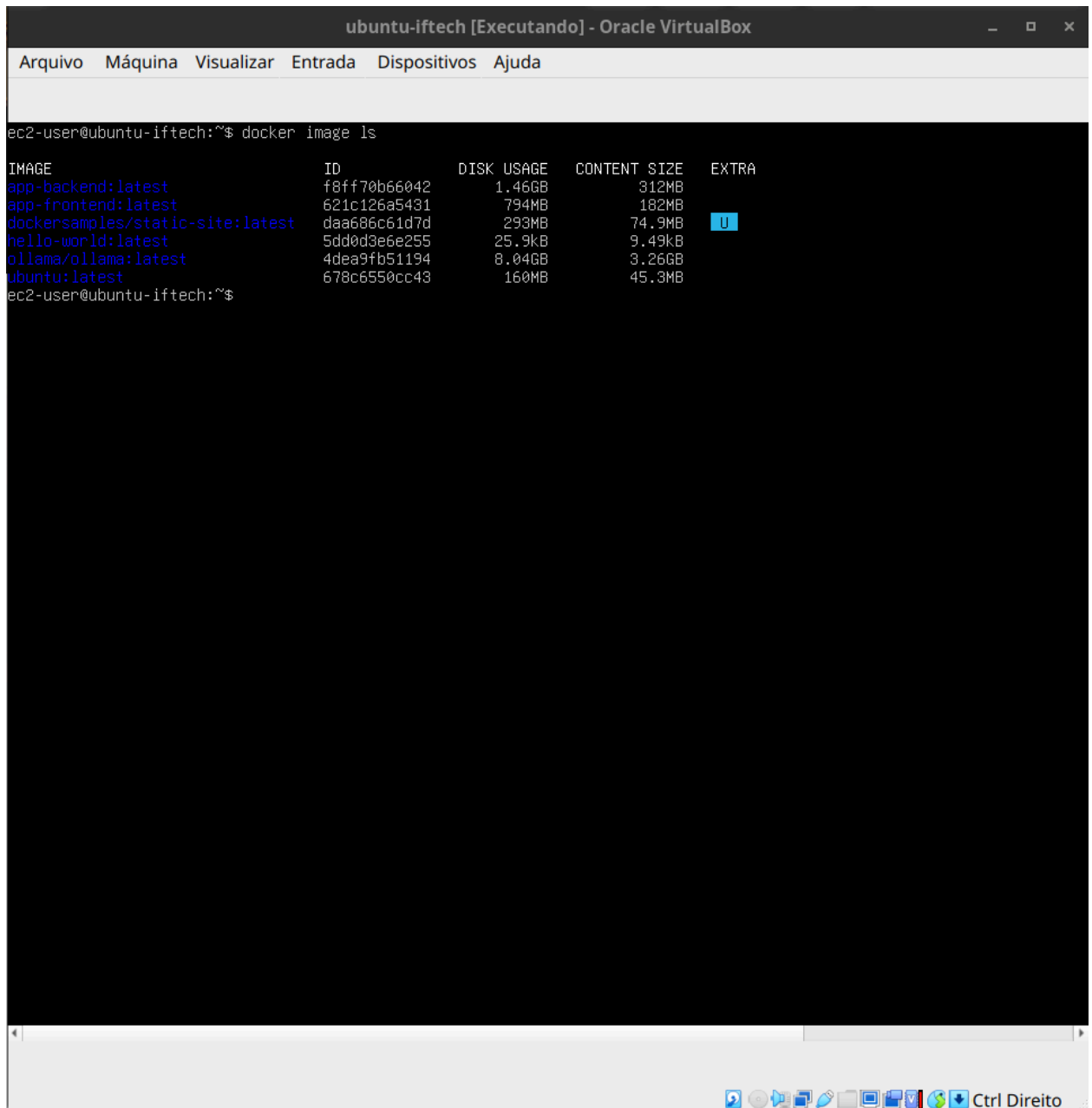
10) Exclui um container com ID dele, por exemplo o ID do container hello-world



```
ubuntu-iftech [Executando] - Oracle VirtualBox
Arquivo  Máquina  Visualizar  Entrada  Dispositivos  Ajuda

ec2-user@ubuntu-iftech:~$ docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
19deb58d6c8d   dockersamples/static-site          "/bin/sh -c 'cd /usr..." 14 minutes ago Up 3 minutes   443/tcp
p_pedantic_varahamihira
66bc3a0835c7   hello-world                         "/hello"                16 minutes ago Exited (0) 16 minutes ago
silly_meninsky
ec2-user@ubuntu-iftech:~$ docker rm 66b
66b
ec2-user@ubuntu-iftech:~$ docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
19deb58d6c8d   dockersamples/static-site          "/bin/sh -c 'cd /usr..." 14 minutes ago Up 4 minutes   443/tcp, 0.0.0.0:808
varahamihira
ec2-user@ubuntu-iftech:~$ _
```

11) Verifica as imagens que o docker já baixou

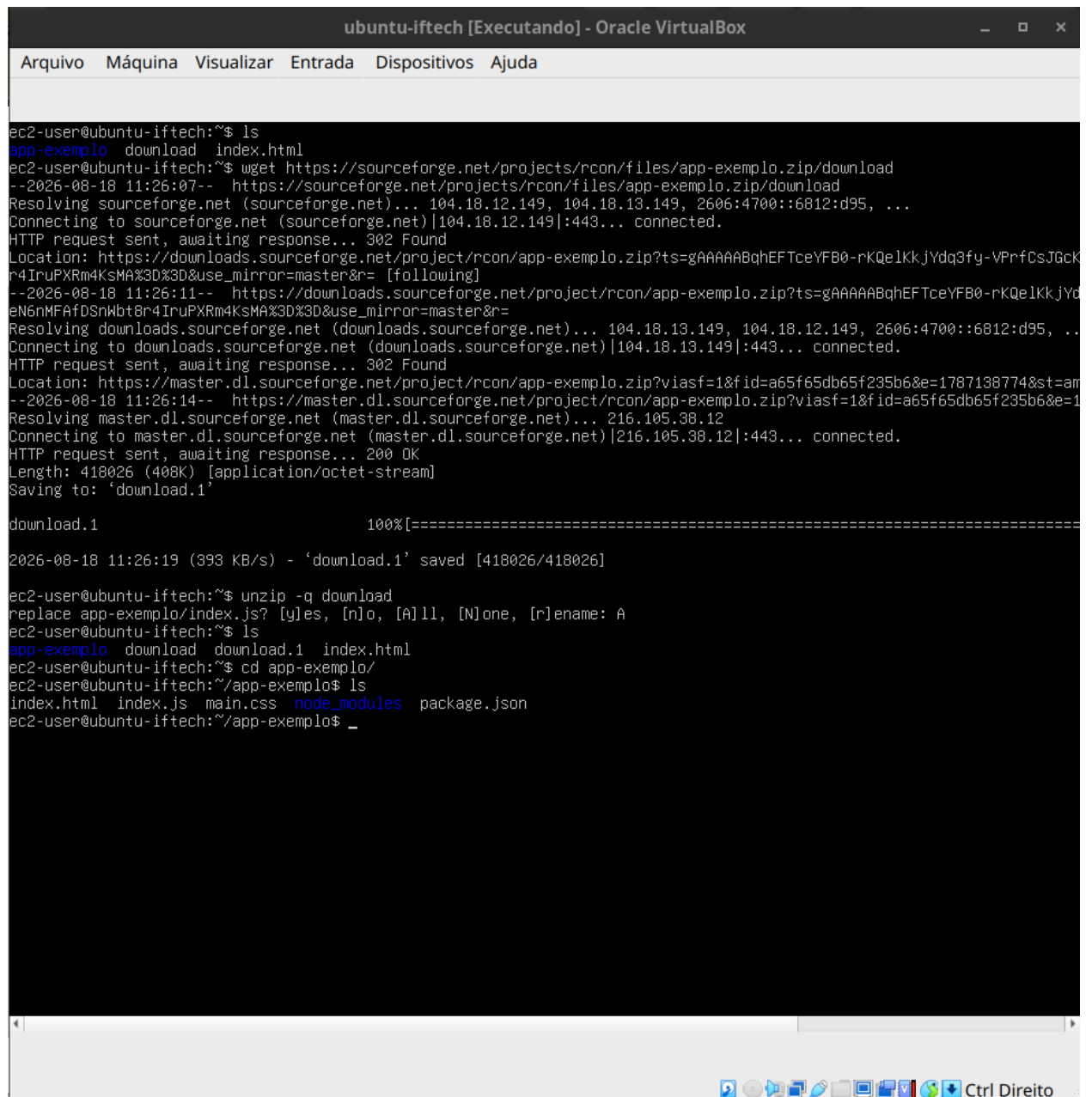


```
ubuntu-iftech [Executando] - Oracle VirtualBox
Arquivo  Máquina  Visualizar  Entrada  Dispositivos  Ajuda

ec2-user@ubuntu-iftech:~$ docker image ls

IMAGE                                ID                                DISK USAGE  CONTENT SIZE  EXTRA
app-backend:latest                  f8ff70b66042                      1.46GB      312MB
app-frontend:latest                 621c126a5431                      794MB       182MB
dockersamples/static-site:latest    daa606c61d7d                      293MB       74.9MB    U
hello-world:latest                  5dd0d3e6e255                      25.9kB      9.49kB
ollama/ollama:latest                4dea9fb51194                      8.04GB      3.26GB
ubuntu:latest                       678c6550cc43                      160MB       45.3MB
ec2-user@ubuntu-iftech:~$
```

- 12) Criar uma imagem docker. Para isso, criar o arquivo dockerfile e depois executar o comando de construção de imagem. Baixar o arquivo app-exemplo.zip



```
ubuntu-iftech [Executando] - Oracle VirtualBox
Arquivo  Máquina  Visualizar  Entrada  Dispositivos  Ajuda

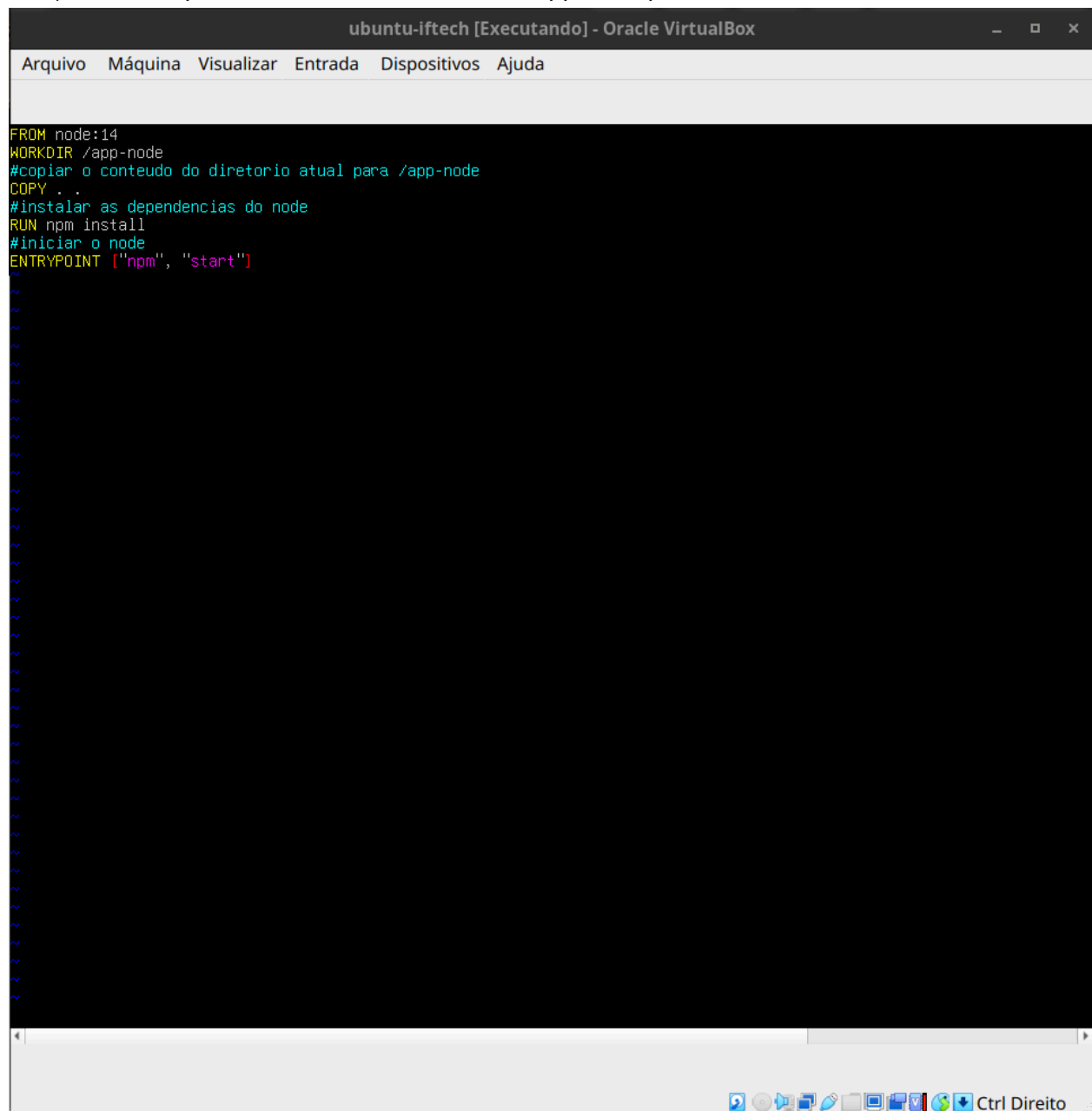
ec2-user@ubuntu-iftech:~$ ls
app-exemplo  download  index.html
ec2-user@ubuntu-iftech:~$ wget https://sourceforge.net/projects/rcon/files/app-exemplo.zip/download
--2026-08-18 11:26:07-- https://sourceforge.net/projects/rcon/files/app-exemplo.zip/download
Resolving sourceforge.net (sourceforge.net)... 104.18.12.149, 104.18.13.149, 2606:4700::6812:d95, ...
Connecting to sourceforge.net (sourceforge.net)|104.18.12.149|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://downloads.sourceforge.net/project/rcon/app-exemplo.zip?ts=gAAAAABqhEFTceYFB0-rKQe1KkJYdq3fy-VPrfCsJGckr4IruPXRM4KsMA%3D%3D&use_mirror=master&r= [following]
--2026-08-18 11:26:11-- https://downloads.sourceforge.net/project/rcon/app-exemplo.zip?ts=gAAAAABqhEFTceYFB0-rKQe1KkJYdq3fy-VPrfCsJGckr4IruPXRM4KsMA%3D%3D&use_mirror=master&r=
Resolving downloads.sourceforge.net (downloads.sourceforge.net)... 104.18.13.149, 104.18.12.149, 2606:4700::6812:d95, ...
Connecting to downloads.sourceforge.net (downloads.sourceforge.net)|104.18.13.149|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://master.dl.sourceforge.net/project/rcon/app-exemplo.zip?viasf=1&fid=a65f65db65f235b6&e=1787138774&st=am
--2026-08-18 11:26:14-- https://master.dl.sourceforge.net/project/rcon/app-exemplo.zip?viasf=1&fid=a65f65db65f235b6&e=1
Resolving master.dl.sourceforge.net (master.dl.sourceforge.net)... 216.105.38.12
Connecting to master.dl.sourceforge.net (master.dl.sourceforge.net)|216.105.38.12|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 418026 (408K) [application/octet-stream]
Saving to: 'download.1'

download.1
100%[=====]

2026-08-18 11:26:19 (393 KB/s) - 'download.1' saved [418026/418026]

ec2-user@ubuntu-iftech:~$ unzip -q download
replace app-exemplo/index.js? [y]es, [n]o, [A]ll, [N]one, [r]ename: A
ec2-user@ubuntu-iftech:~$ ls
app-exemplo  download  download.1  index.html
ec2-user@ubuntu-iftech:~$ cd app-exemplo/
ec2-user@ubuntu-iftech:~/app-exemplo$ ls
index.html  index.js  main.css  node_modules  package.json
ec2-user@ubuntu-iftech:~/app-exemplo$ _
```

12.1) Criar o arquivo Dockerfile no diretório do app-exemplo



The screenshot shows a terminal window titled "ubuntu-iftech [Executando] - Oracle VirtualBox". The terminal has a menu bar with "Arquivo", "Máquina", "Visualizar", "Entrada", "Dispositivos", and "Ajuda". The terminal content is as follows:

```
FROM node:14
WORKDIR /app-node
#copiar o conteúdo do diretório atual para /app-node
COPY . .
#instalar as dependências do node
RUN npm install
#iniciar o node
ENTRYPOINT ["npm", "start"]
```

The terminal shows a series of blue tilde characters (~) indicating that the command prompt is active and ready for input. At the bottom of the window, there is a taskbar with various icons and the text "Ctrl Direito".

12.2) Criar a imagem usando o dockerfile com nome rcon , considerando o workdir, sendo a versao1.0 e considerando os arquivo do diretório atual (.)

```
ubuntu-iftech [Executando] - Oracle VirtualBox
Arquivo  Máquina  Visualizar  Entrada  Dispositivos  Ajuda

ec2-user@ubuntu-iftech:~/app-exemplo$ docker build -t rcon/app-node:1.0 .
[+] Building 44.0s (9/9) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 226B
=> [internal] load metadata for docker.io/library/node:14
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/4] FROM docker.io/library/node:14@sha256:a158d3b9b4e3fa813fa6c8c590b8f0a860e015ad4e59bbce5744d2f6fd8461aa
=> => resolve docker.io/library/node:14@sha256:a158d3b9b4e3fa813fa6c8c590b8f0a860e015ad4e59bbce5744d2f6fd8461aa
=> => sha256:0d27a8e861329007574c6766fba946d48e20d2c8e964e873de352603f22c4ceb 450B / 450B
=> => sha256:0c8cc2f24a4dcb64e602e086fc9446b0a541e8acd9ad72d2e90df3ba22f158b3 2.29MB / 2.29MB
=> => sha256:5f32ed3c3f278edda4fc571c880b5277355a29ae8f52b52cdf865f058378a590 35.24MB / 35.24MB
=> => sha256:6f51ee005deac0d99898e41b8ce60ebf250ebe1a31a0b03f613aec6bbcb9b83d8 4.19kB / 4.19kB
=> => sha256:1de76e268b103d05fa8960e0f77951ff54b912b63429c34f5d6adfd09f5f9ee2 51.88MB / 51.88MB
=> => sha256:d9a8df5894511ce28a05e2925a75e8a4acbd0634c39ad734fd8ba8e23d1b1569 191.85MB / 191.85MB
=> => sha256:3d2201bd995ccc12851a50820de03d34a17011dcbb9ac9fdf3a50c952cbb131 10.00MB / 10.00MB
=> => sha256:b253aeafeaa7e0671bb60008df01de101a38a045ff7bc656e3b0fbfc7c05cca5 7.86MB / 7.86MB
=> => sha256:2ff1d7c41c74a25258bfa6f0b8adb0a727f84518f55f65ca845ebc747976c408 50.45MB / 50.45MB
=> => extracting sha256:2ff1d7c41c74a25258bfa6f0b8adb0a727f84518f55f65ca845ebc747976c408
=> => extracting sha256:b253aeafeaa7e0671bb60008df01de101a38a045ff7bc656e3b0fbfc7c05cca5
=> => extracting sha256:3d2201bd995ccc12851a50820de03d34a17011dcbb9ac9fdf3a50c952cbb131
=> => extracting sha256:1de76e268b103d05fa8960e0f77951ff54b912b63429c34f5d6adfd09f5f9ee2
=> => extracting sha256:d9a8df5894511ce28a05e2925a75e8a4acbd0634c39ad734fd8ba8e23d1b1569
=> => extracting sha256:6f51ee005deac0d99898e41b8ce60ebf250ebe1a31a0b03f613aec6bbcb9b83d8
=> => extracting sha256:5f32ed3c3f278edda4fc571c880b5277355a29ae8f52b52cdf865f058378a590
=> => extracting sha256:0c8cc2f24a4dcb64e602e086fc9446b0a541e8acd9ad72d2e90df3ba22f158b3
=> => extracting sha256:0d27a8e861329007574c6766fba946d48e20d2c8e964e873de352603f22c4ceb
=> [internal] load build context
=> => transferring context: 1.15MB
=> [2/4] WORKDIR /app-node
=> [3/4] COPY . .
=> [4/4] RUN npm install
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:c6f72312be893f5049299f760585397d9db1f46a777960dd094f34cfe9b7c7cc
=> => exporting config sha256:2583d215ed30aa044df9675c8326e9decad02acbbcae05a42ada47fdf4eee404
=> => exporting attestation manifest sha256:57cfd02c0dbf3822b4b6f8c6c9ce798c9a9aa7205c9019fc6bed1bdf66af8da1
=> => exporting manifest list sha256:fc9531a8c0a7931d960b26dd23e941feb34b020d37cc9bf4bc219d9746610ab0
=> => naming to docker.io/rcon/app-node:1.0
=> => unpacking to docker.io/rcon/app-node:1.0
ec2-user@ubuntu-iftech:~/app-exemplo$
```

12.3) Imagem criada

```
ubuntu-iftech [Executando] - Oracle VirtualBox
Arquivo  Máquina  Visualizar  Entrada  Dispositivos  Ajuda

ec2-user@ubuntu-iftech:~/app-exemplo$ docker image ls

IMAGE                                ID                                DISK USAGE  CONTENT SIZE  EXTRA
app-backend:latest                  f8ff70b66042                      1.46GB      312MB
app-frontend:latest                 621c126a5431                      794MB       182MB
dockersamples/static-site:latest    daa686c61d7d                      293MB       74.9MB    U
hello-world:latest                  5dd0d3e6e255                      25.9kB      9.49kB
ollama/ollama:latest                4dea9fb51194                      8.04GB      3.26GB
rcon/app-node:1.0                   fc9531a8c0a7                      1.35GB      350MB
ubuntu:latest                       678c6550cc43                      160MB       45.3MB

ec2-user@ubuntu-iftech:~/app-exemplo$ _
```



12.4) Executar um container com a imagem criada

```
ubuntu-iftech [Executando] - Oracle VirtualBox
Arquivo  Máquina  Visualizar  Entrada  Dispositivos  Ajuda

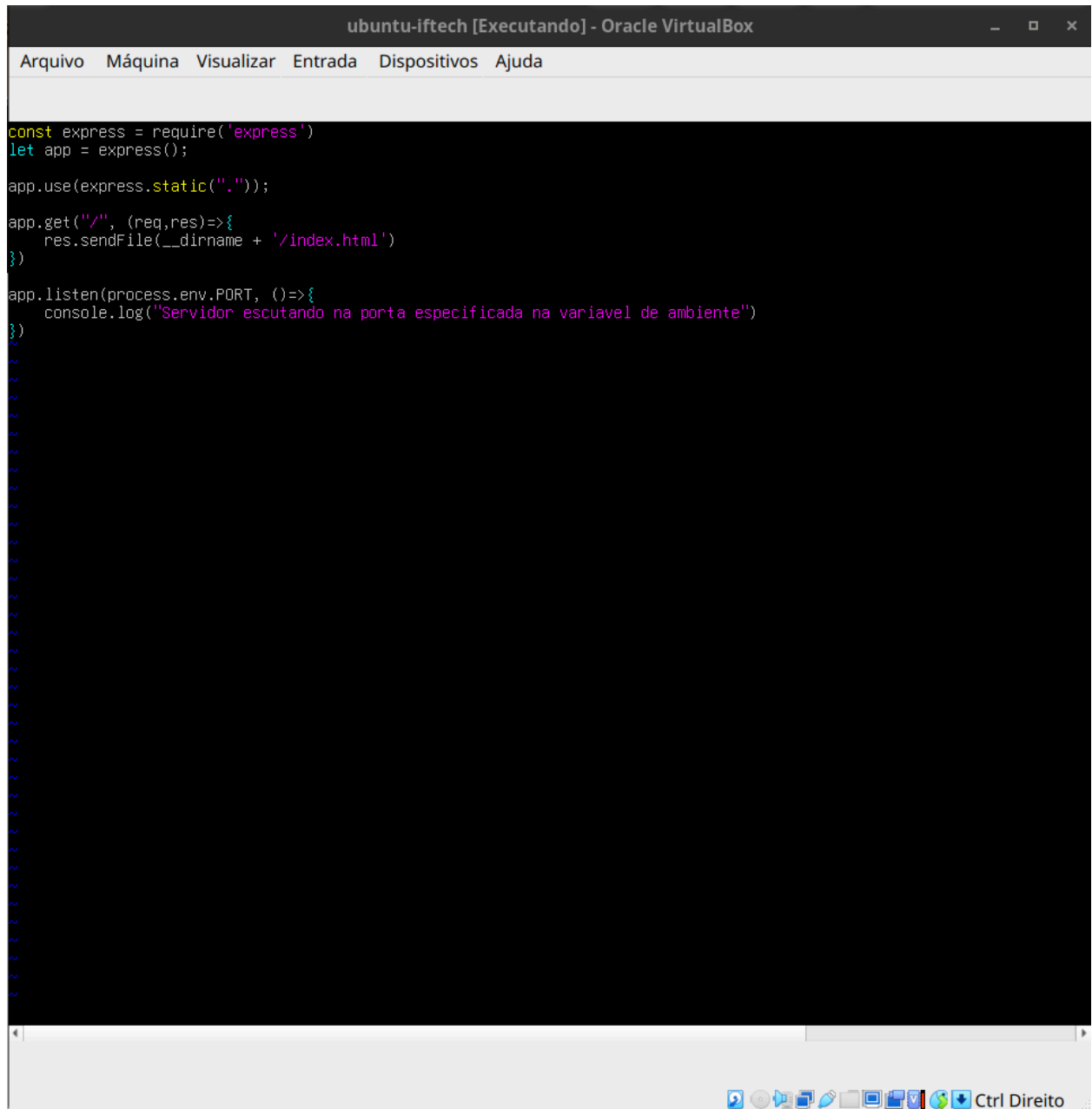
ec2-user@ubuntu-iftech:~/app-exemplo$ docker run -d -p 8082:3000 ncon/app-node:1.0
2ca44db5f8ea657166351e71e1c543c53c419419336c6f2015239f97f16a669b
ec2-user@ubuntu-iftech:~/app-exemplo$ ls
Dockerfile  index.html  index.js  main.css  node_modules  package.json
ec2-user@ubuntu-iftech:~/app-exemplo$ cd
ec2-user@ubuntu-iftech:~$ rm index.html
ec2-user@ubuntu-iftech:~$ wget http://localhost:8082/
--2026-08-18 11:36:03--  http://localhost:8082/
Resolving localhost (localhost)... ::1, 127.0.0.1
Connecting to localhost (localhost)|::1|:8082... connected.
HTTP request sent, awaiting response... 200 OK
Length: 288 [text/html]
Saving to: 'index.html'

index.html                                     100%[=====]

2026-08-18 11:36:03 (27,1 MB/s) - 'index.html' saved [288/288]

ec2-user@ubuntu-iftech:~$ head -n 15 index.html
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width">
  <title>Node <3</title>
  <link rel="stylesheet" href="main.css">
</head>
<body>
  <h1>Executei um servidor com Node em um container!</h1>
</body>
</html>
ec2-user@ubuntu-iftech:~$ _
```


13.1) Criar o arquivo Dockerfile no diretório do app-exemplo, alterando o arquivo para ler uma variável de ambiente configurada no dockerfile



The image shows a terminal window titled "ubuntu-iftech [Executando] - Oracle VirtualBox". The terminal displays the following JavaScript code for an Express server:

```
const express = require('express')
let app = express();

app.use(express.static("."));

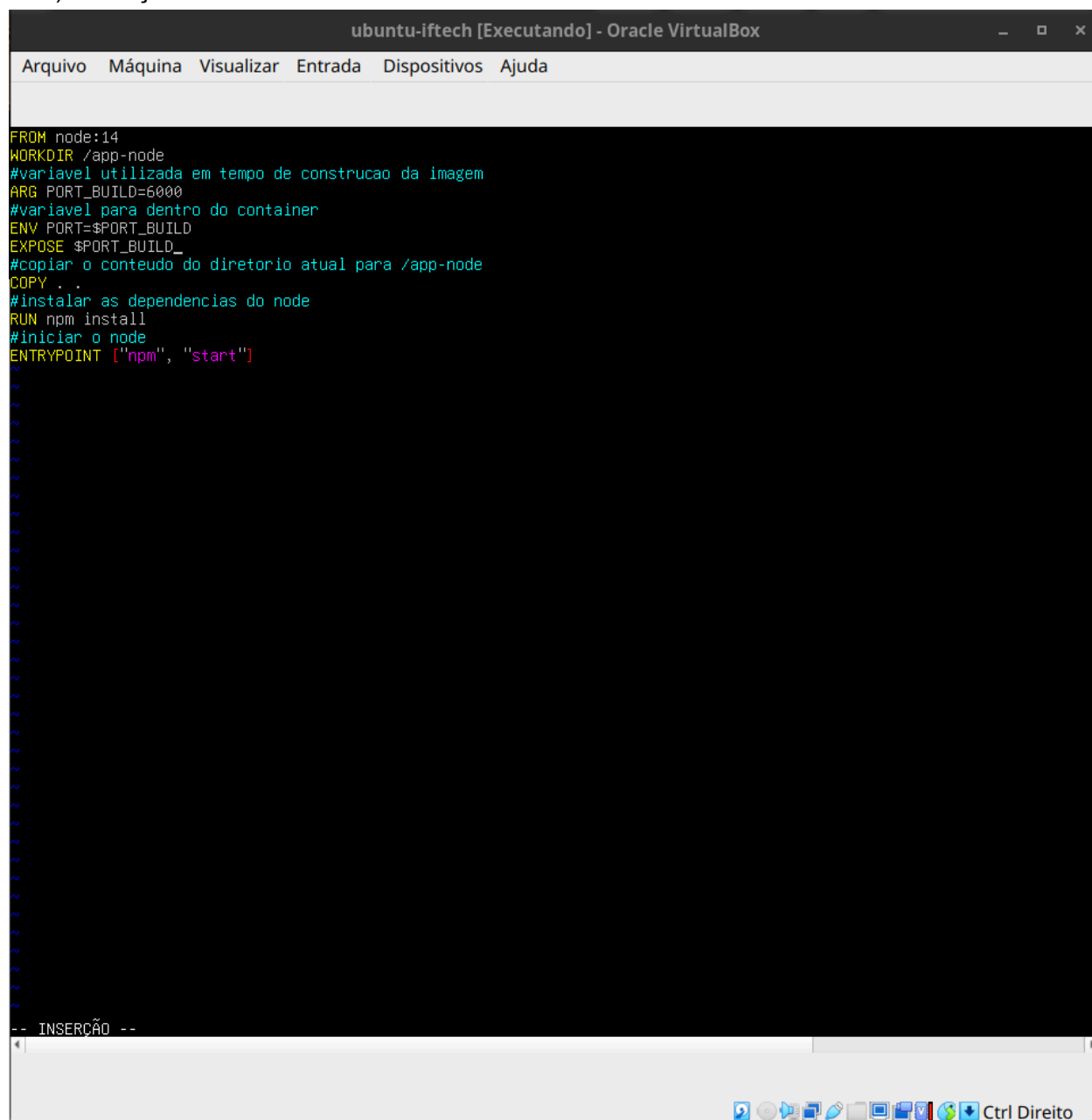
app.get("/", (req,res)=>{
  res.sendFile(__dirname + '/index.html')
})

app.listen(process.env.PORT, ()=>{
  console.log("Servidor escutando na porta especificada na variavel de ambiente")
})
```

The code is color-coded: `const` is blue, `express` is green, `require` is red, `let` is blue, `app` is green, `use` is red, `static` is green, `get` is red, `res.sendFile` is red, `listen` is red, `process.env` is green, `PORT` is green, `console.log` is red, and the string `"Servidor escutando na porta especificada na variavel de ambiente"` is pink. The terminal also shows a series of tilde characters (~) representing the output of the `listen` function.

The window has a menu bar with "Arquivo", "Máquina", "Visualizar", "Entrada", "Dispositivos", and "Ajuda". The bottom status bar shows system icons and the text "Ctrl Direito".

13.2) Alteração do Dockerfile



```
FROM node:14
WORKDIR /app-node
#variavel utilizada em tempo de construcao da imagem
ARG PORT_BUILD=6000
#variavel para dentro do container
ENV PORT=$PORT_BUILD
EXPOSE $PORT_BUILD_
#copiar o conteudo do diretorio atual para /app-node
COPY . .
#instalar as dependencias do node
RUN npm install
#iniciar o node
ENTRYPOINT ["npm", "start"]

-- INSCRIÇÃO --
```

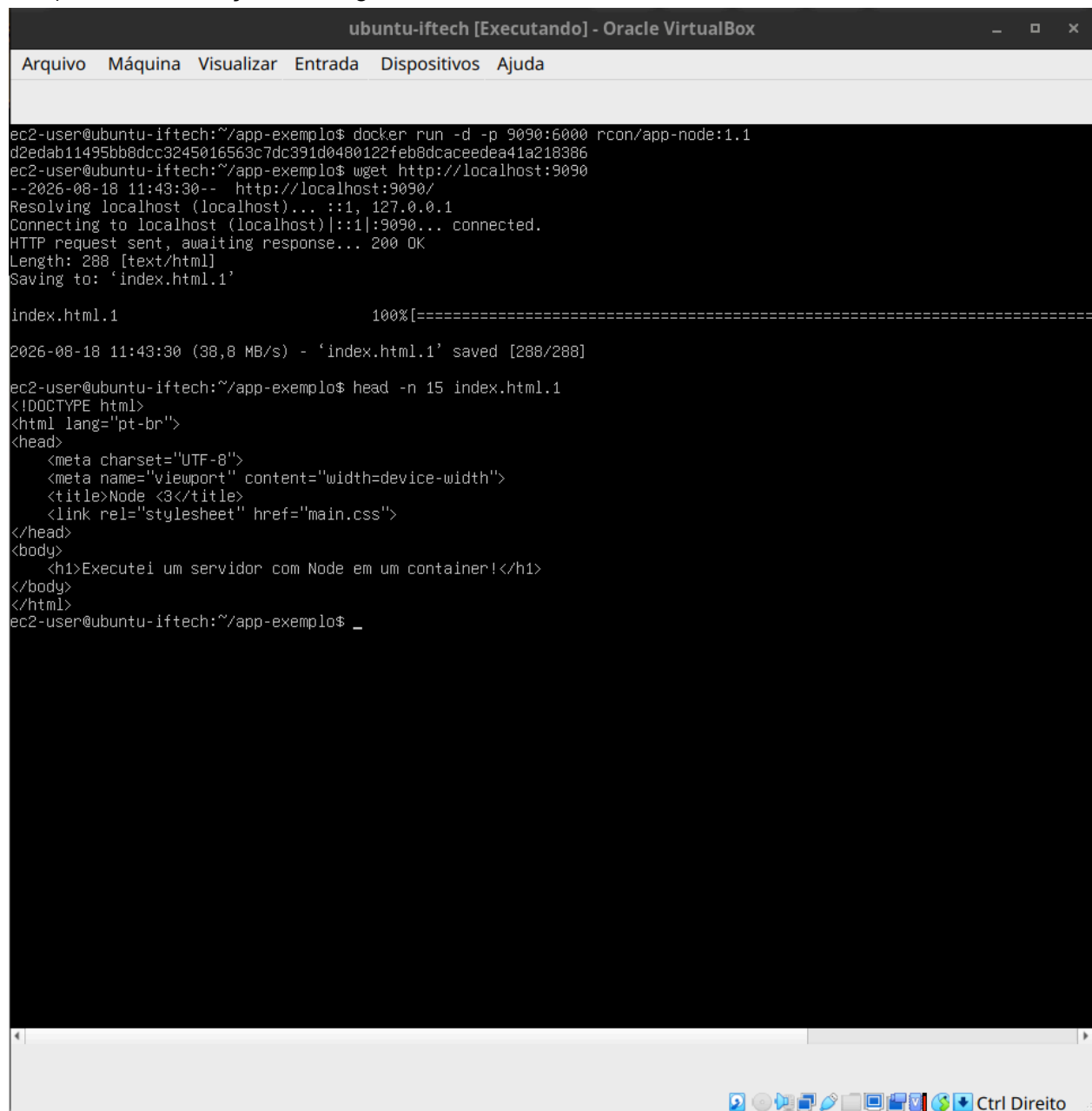
Ctrl Direito

13.3) criar a nova imagem com a versão 1.1

```
ubuntu-iftech [Executando] - Oracle VirtualBox
Arquivo  Máquina  Visualizar  Entrada  Dispositivos  Ajuda

ec2-user@ubuntu-iftech:~/app-exemplo$ docker build -t rcon/app-node:1.1 .
[+] Building 18.6s (2/2) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 374B
=> ERROR [internal] load metadata for docker.io/library/node:14
-----
> [internal] load metadata for docker.io/library/node:14:
-----
Dockerfile:1
-----
1 | >>> FROM node:14
2 |     WORKDIR /app-node
3 |     #variavel utilizada em tempo de construcao da imagem
-----
ERROR: failed to build: failed to solve: node:14: failed to resolve source metadata for docker.io/library/node:14: failed
1.docker.io/v2/library/node/manifests/14": dial tcp [2600:1f18:2148:bc02:cb87:c157:290f:ec7d]:443: connect: network is u
ec2-user@ubuntu-iftech:~/app-exemplo$ docker build -t rcon/app-node:1.1 .
[+] Building 10.7s (9/9) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 374B
=> [internal] load metadata for docker.io/library/node:14
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/4] FROM docker.io/library/node:14@sha256:a158d3b9b4e3fa813fa6c8c590b8f0a860e015ad4e59bbce5744d2f6fd8461aa
=> => resolve docker.io/library/node:14@sha256:a158d3b9b4e3fa813fa6c8c590b8f0a860e015ad4e59bbce5744d2f6fd8461aa
=> [internal] load build context
=> => transferring context: 17.01kB
=> CACHED [2/4] WORKDIR /app-node
=> [3/4] COPY . .
=> [4/4] RUN npm install
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:d6f126ad43c3a8929f8224ad87287b56dcfc8e60e3e8b071972ba99fb074a9a0
=> => exporting config sha256:3045debea839c81e222773b5f12fc2421699629566ee38bda6c491ede93726e
=> => exporting attestation manifest sha256:7a92e810331aab1c3f7774e48ff795939125a79dd5bfc1f4f5d9e96fa8e8145
=> => exporting manifest list sha256:7352c25af5707ca905a5739d9cd937cd1e4cd0d4c875dce15bedfad74b532694
=> => naming to docker.io/rcon/app-node:1.1
=> => unpacking to docker.io/rcon/app-node:1.1
ec2-user@ubuntu-iftech:~/app-exemplo$ _
```

13.4) testar a execução da imagem criada



```
ubuntu-iftech [Executando] - Oracle VirtualBox
Arquivo  Máquina  Visualizar  Entrada  Dispositivos  Ajuda

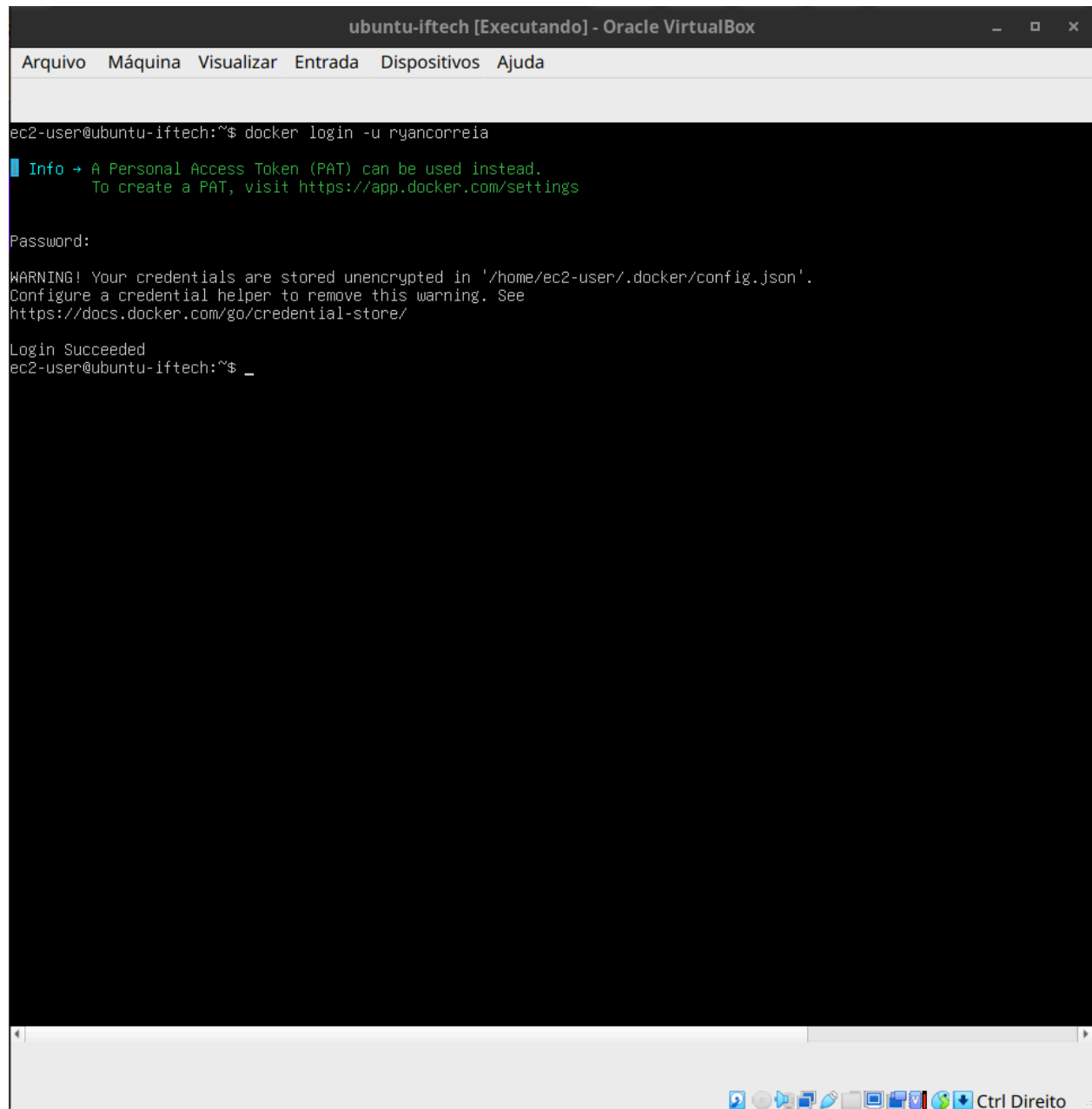
ec2-user@ubuntu-iftech:~/app-exemplo$ docker run -d -p 9090:6000 ncon/app-node:1.1
d2edab11495bb8dccc3245016563c7dc391d0480122feb8dcaceede41a218386
ec2-user@ubuntu-iftech:~/app-exemplo$ wget http://localhost:9090
--2026-08-18 11:43:30--  http://localhost:9090/
Resolving localhost (localhost)... ::1, 127.0.0.1
Connecting to localhost (localhost)|::1|:9090... connected.
HTTP request sent, awaiting response... 200 OK
Length: 288 [text/html]
Saving to: 'index.html.1'

index.html.1
100%[=====]

2026-08-18 11:43:30 (38,8 MB/s) - 'index.html.1' saved [288/288]

ec2-user@ubuntu-iftech:~/app-exemplo$ head -n 15 index.html.1
<!DOCTYPE html>
<html lang="pt-br">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width">
  <title>Node <3</title>
  <link rel="stylesheet" href="main.css">
</head>
<body>
  <h1>Executei um servidor com Node em um container!</h1>
</body>
</html>
ec2-user@ubuntu-iftech:~/app-exemplo$ _
```

14) Autenticar a conta ao dockerhub



The screenshot shows a terminal window titled "ubuntu-iftech [Executando] - Oracle VirtualBox". The terminal displays the command `docker login -u ryancorreia` and its output. The output includes an informational message about Personal Access Tokens (PAT), a password prompt, a warning about unencrypted credentials, and a successful login confirmation.

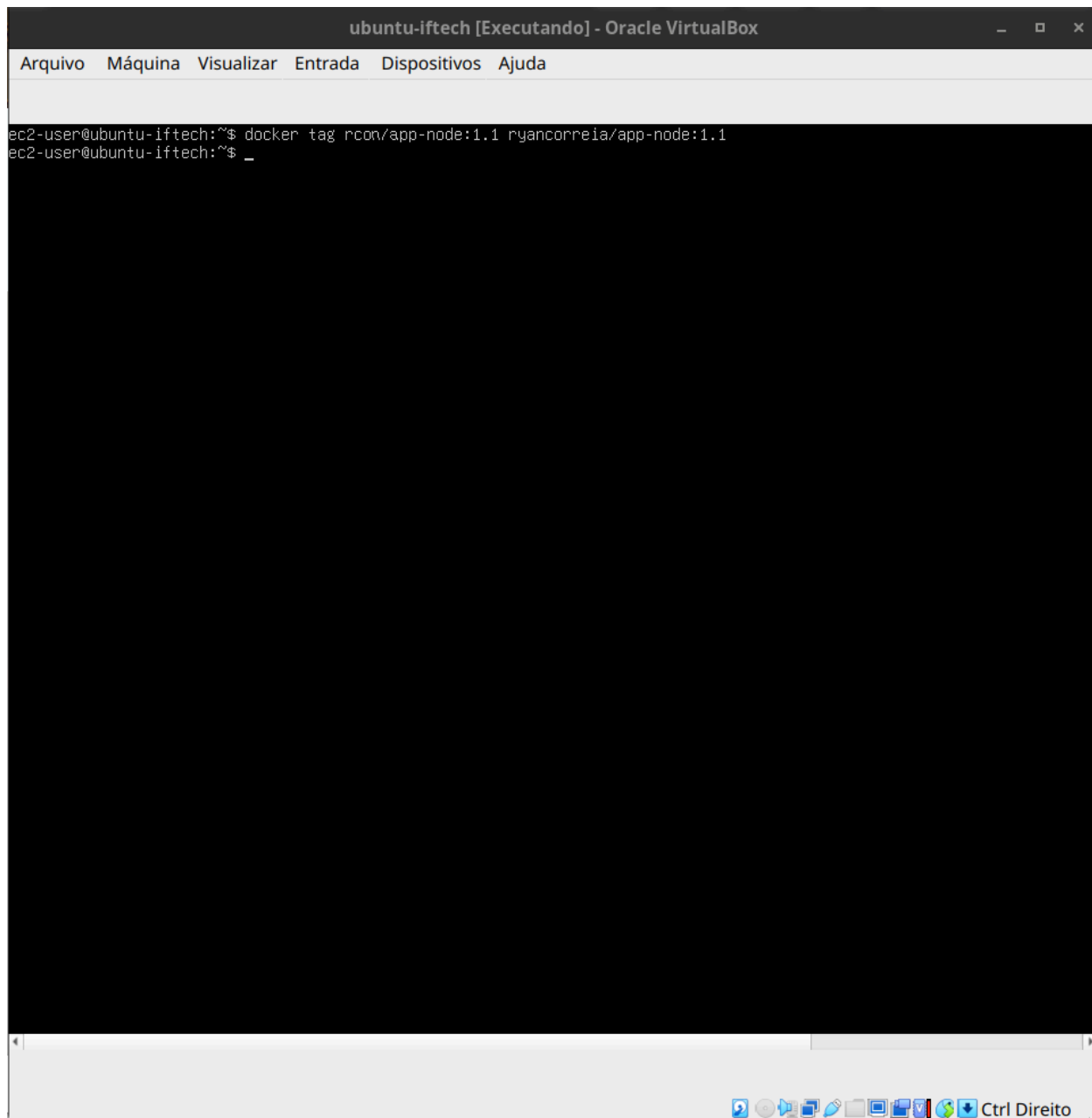
```
ec2-user@ubuntu-iftech:~$ docker login -u ryancorreia
Info → A Personal Access Token (PAT) can be used instead.
       To create a PAT, visit https://app.docker.com/settings

Password:

WARNING! Your credentials are stored unencrypted in '/home/ec2-user/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
ec2-user@ubuntu-iftech:~$ _
```

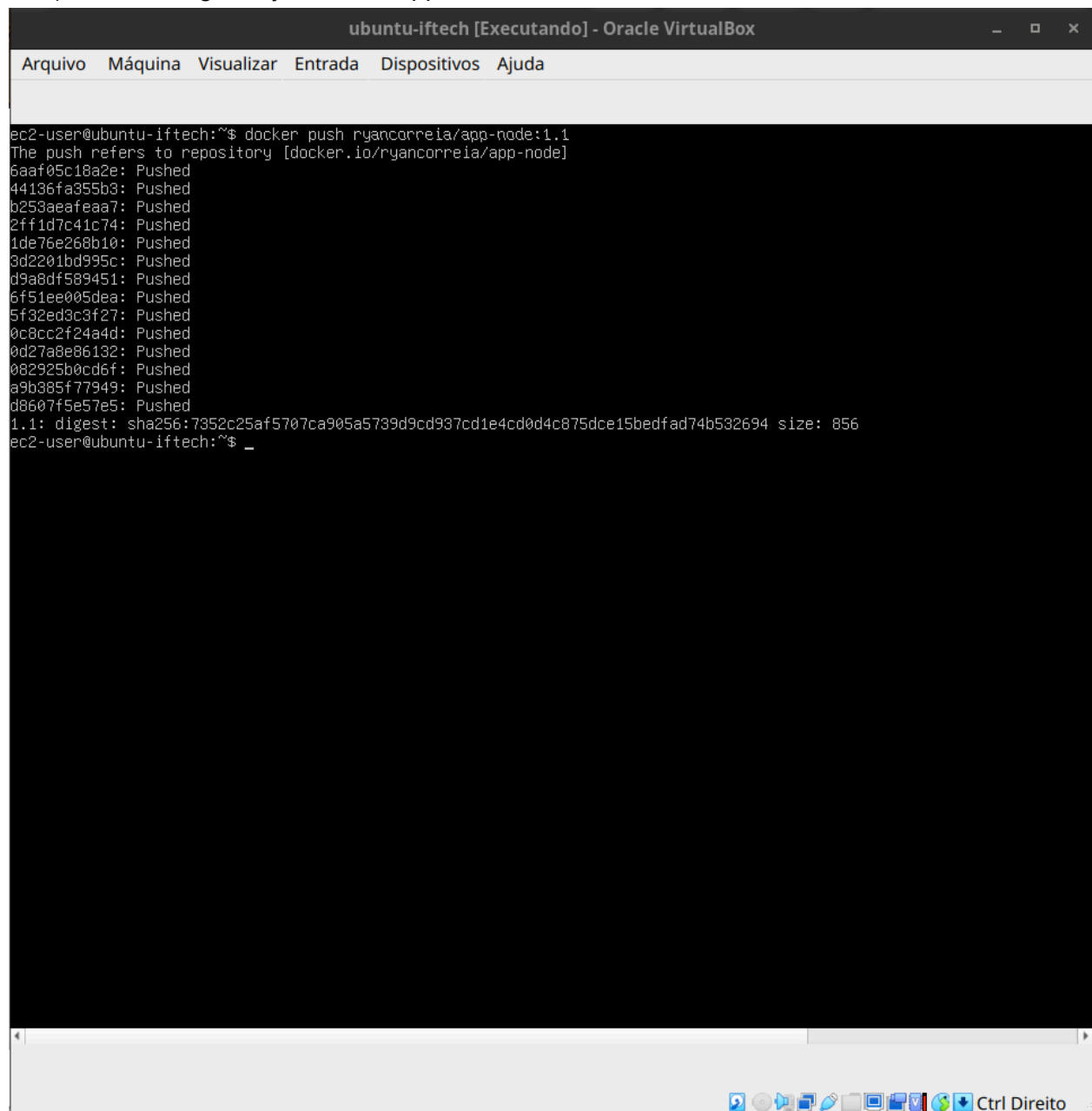
14.3) criar uma tag para a imagem criada rcon/app-node:1.1 considerando o nome da minha conta no dockerhub



The screenshot shows a terminal window titled "ubuntu-iftech [Executando] - Oracle VirtualBox". The window has a menu bar with "Arquivo", "Máquina", "Visualizar", "Entrada", "Dispositivos", and "Ajuda". The terminal content shows the user "ec2-user@ubuntu-iftech" executing the command "docker tag rcon/app-node:1.1 ryanconreia/app-node:1.1". The prompt changes from "~\$" to "~\$ _" after the command is entered. The terminal background is black, and the text is white. At the bottom of the window, there is a taskbar with various icons and the text "Ctrl Direito".

```
ec2-user@ubuntu-iftech:~$ docker tag rcon/app-node:1.1 ryanconreia/app-node:1.1
ec2-user@ubuntu-iftech:~$ _
```

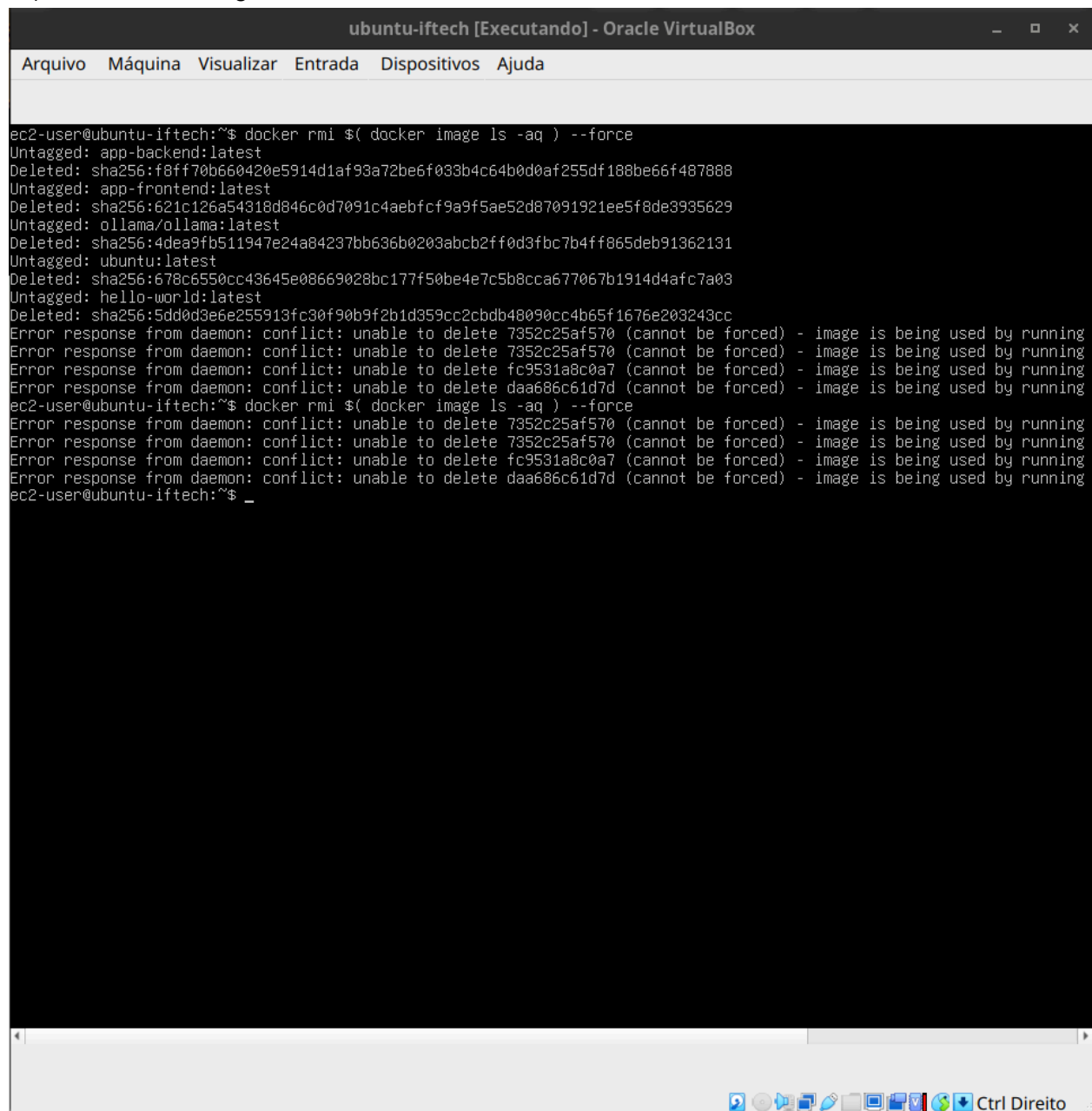
14.4) Subir a imagem: ryanorreia/app-node:1.1



The screenshot shows a terminal window titled "ubuntu-iftech [Executando] - Oracle VirtualBox". The terminal displays the command `docker push ryanorreia/app-node:1.1` and its output. The output indicates that the image was successfully pushed to the repository `docker.io/ryanorreia/app-node`. The terminal also shows the digest and size of the image.

```
ec2-user@ubuntu-iftech:~$ docker push ryanorreia/app-node:1.1
The push refers to repository [docker.io/ryanorreia/app-node]
6aaf05c18a2e: Pushed
44136fa355b3: Pushed
b253aafeaa7: Pushed
2ff1d7c41c74: Pushed
1de76e268b10: Pushed
3d2201bd995c: Pushed
d9a8df589451: Pushed
6f51ee005dea: Pushed
5f32ed3c3f27: Pushed
0c8cc2f24a4d: Pushed
0d27a8e86132: Pushed
082925b0cd6f: Pushed
a9b385f77949: Pushed
d8607f5e57e5: Pushed
1.1: digest: sha256:7352c25af5707ca905a5739d9cd937cd1e4cd0d4c875dce15bedfad74b532694 size: 856
ec2-user@ubuntu-iftech:~$ _
```

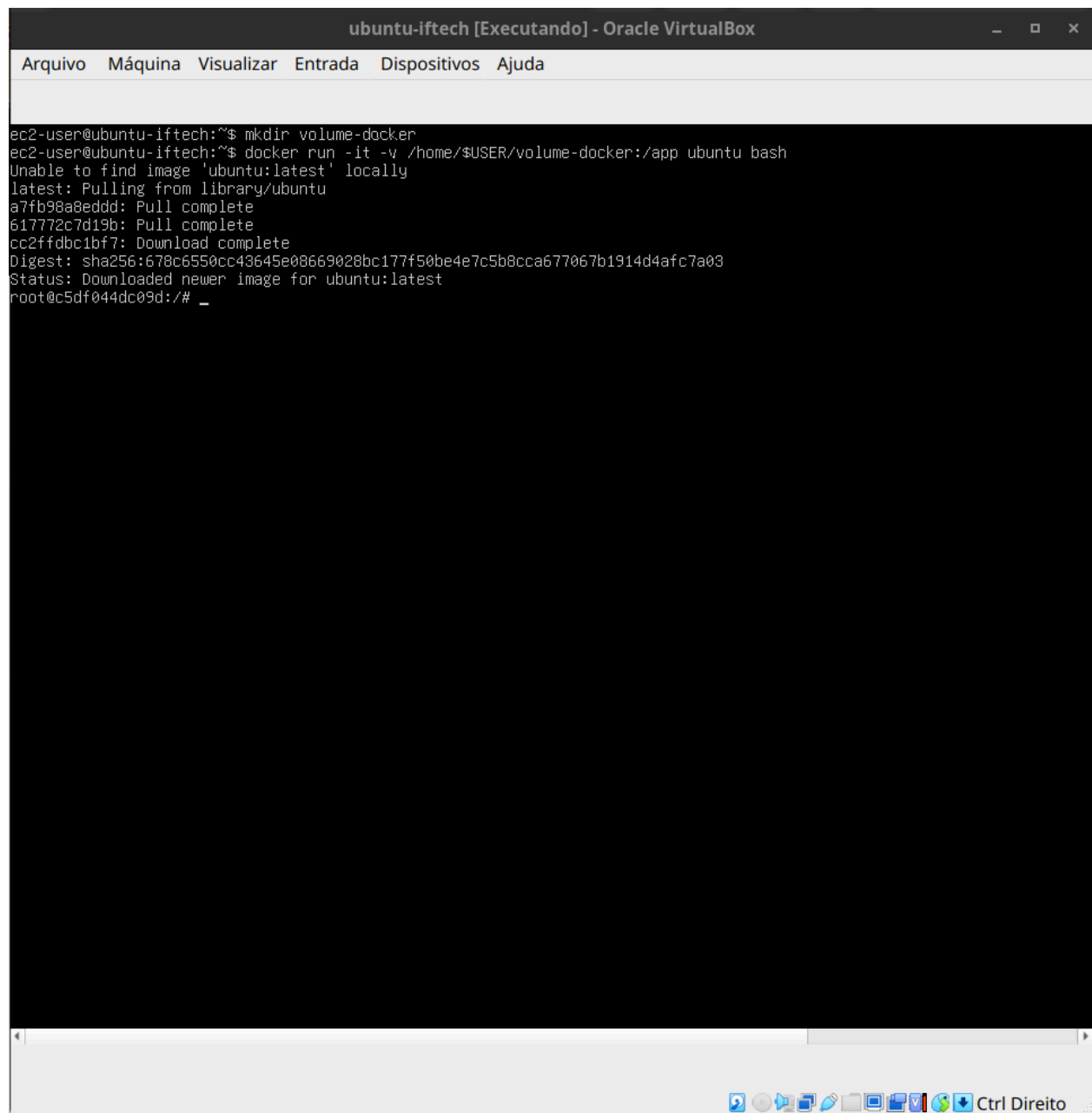
15) Remover as imagens dos containers



```
ubuntu-iftech [Executando] - Oracle VirtualBox
Arquivo  Máquina  Visualizar  Entrada  Dispositivos  Ajuda

ec2-user@ubuntu-iftech:~$ docker rmi $(docker image ls -aq) --force
Untagged: app-backend:latest
Deleted: sha256:f8ff70b660420e5914d1af93a72be6f033b4c64b0d0af255df188be66f487888
Untagged: app-frontend:latest
Deleted: sha256:621c126a54318d846c0d7091c4aebfcf9a9f5ae52d87091921ee5f8de3935629
Untagged: ollama/ollama:latest
Deleted: sha256:4dea9fb511947e24a84237bb636b0203abcb2ff0d3fbc7b4ff865deb91362131
Untagged: ubuntu:latest
Deleted: sha256:678c6550cc43645e08669028bc177f50be4e7c5b8cca677067b1914d4afc7a03
Untagged: hello-world:latest
Deleted: sha256:5dd0d3e6e255913fc30f90b9f2b1d359cc2cbdb48090cc4b65f1676e203243cc
Error response from daemon: conflict: unable to delete 7352c25af570 (cannot be forced) - image is being used by running
Error response from daemon: conflict: unable to delete 7352c25af570 (cannot be forced) - image is being used by running
Error response from daemon: conflict: unable to delete fc9531a8c0a7 (cannot be forced) - image is being used by running
Error response from daemon: conflict: unable to delete daa686c61d7d (cannot be forced) - image is being used by running
ec2-user@ubuntu-iftech:~$ docker rmi $(docker image ls -aq) --force
Error response from daemon: conflict: unable to delete 7352c25af570 (cannot be forced) - image is being used by running
Error response from daemon: conflict: unable to delete 7352c25af570 (cannot be forced) - image is being used by running
Error response from daemon: conflict: unable to delete fc9531a8c0a7 (cannot be forced) - image is being used by running
Error response from daemon: conflict: unable to delete daa686c61d7d (cannot be forced) - image is being used by running
ec2-user@ubuntu-iftech:~$ _
```

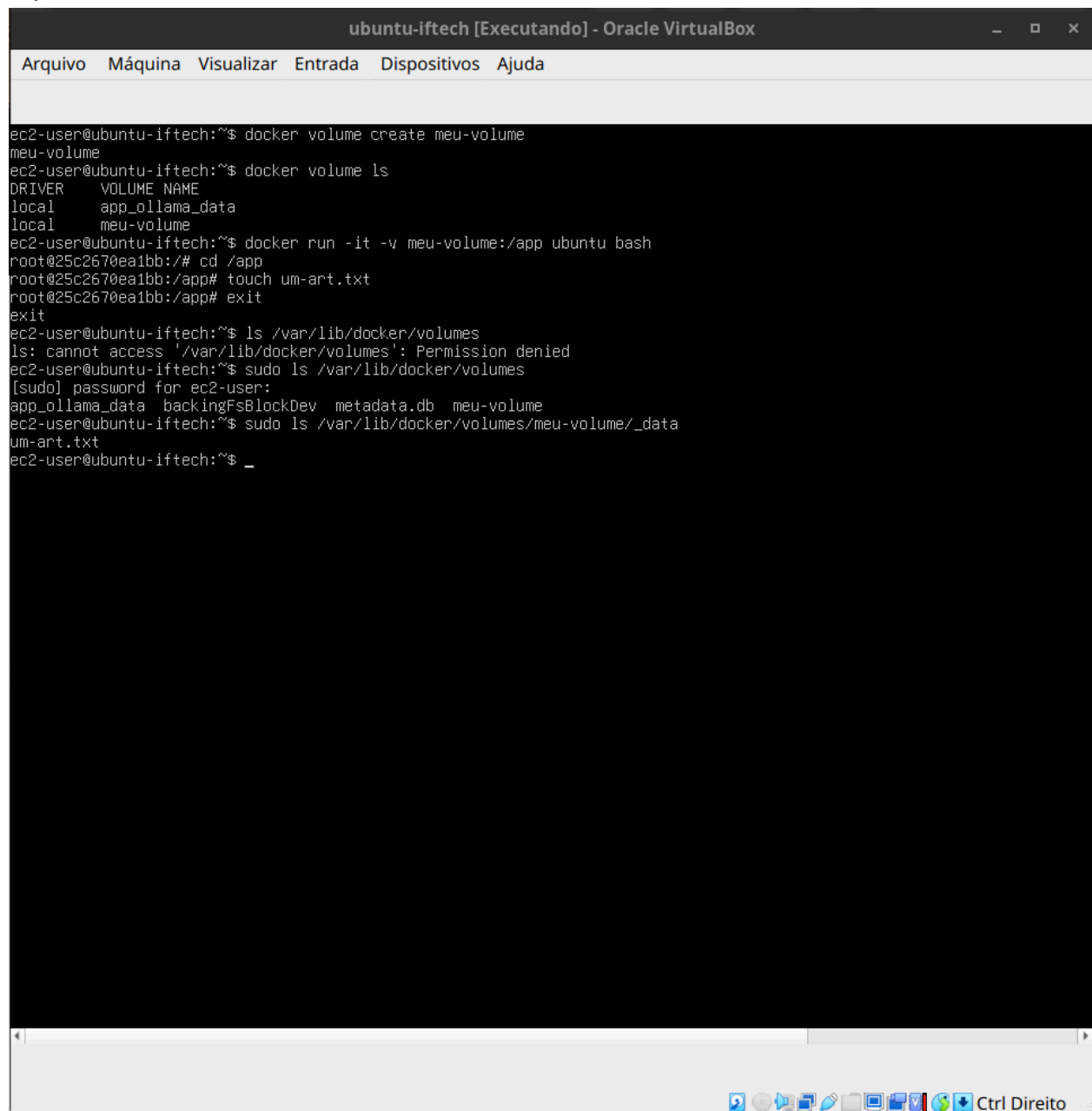

16) executar um container e salvar arquivos de forma que a máquina host veja os arquivos em /home/\$USER/volume-docker e este esteja vinculado a /app no docker
cd /home/\$USER/



```
ubuntu-iftech [Executando] - Oracle VirtualBox
Arquivo  Máquina  Visualizar  Entrada  Dispositivos  Ajuda

ec2-user@ubuntu-iftech:~$ mkdir volume-docker
ec2-user@ubuntu-iftech:~$ docker run -it -v /home/$USER/volume-docker:/app ubuntu bash
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
a7fb98a8eddd: Pull complete
617772c7d19b: Pull complete
cc2ffdbc1bf7: Download complete
Digest: sha256:678c6550cc43645e08669028bc177f50be4e7c5b8cca677067b1914d4afc7a03
Status: Downloaded newer image for ubuntu:latest
root@c5df044dc09d:/# _
```

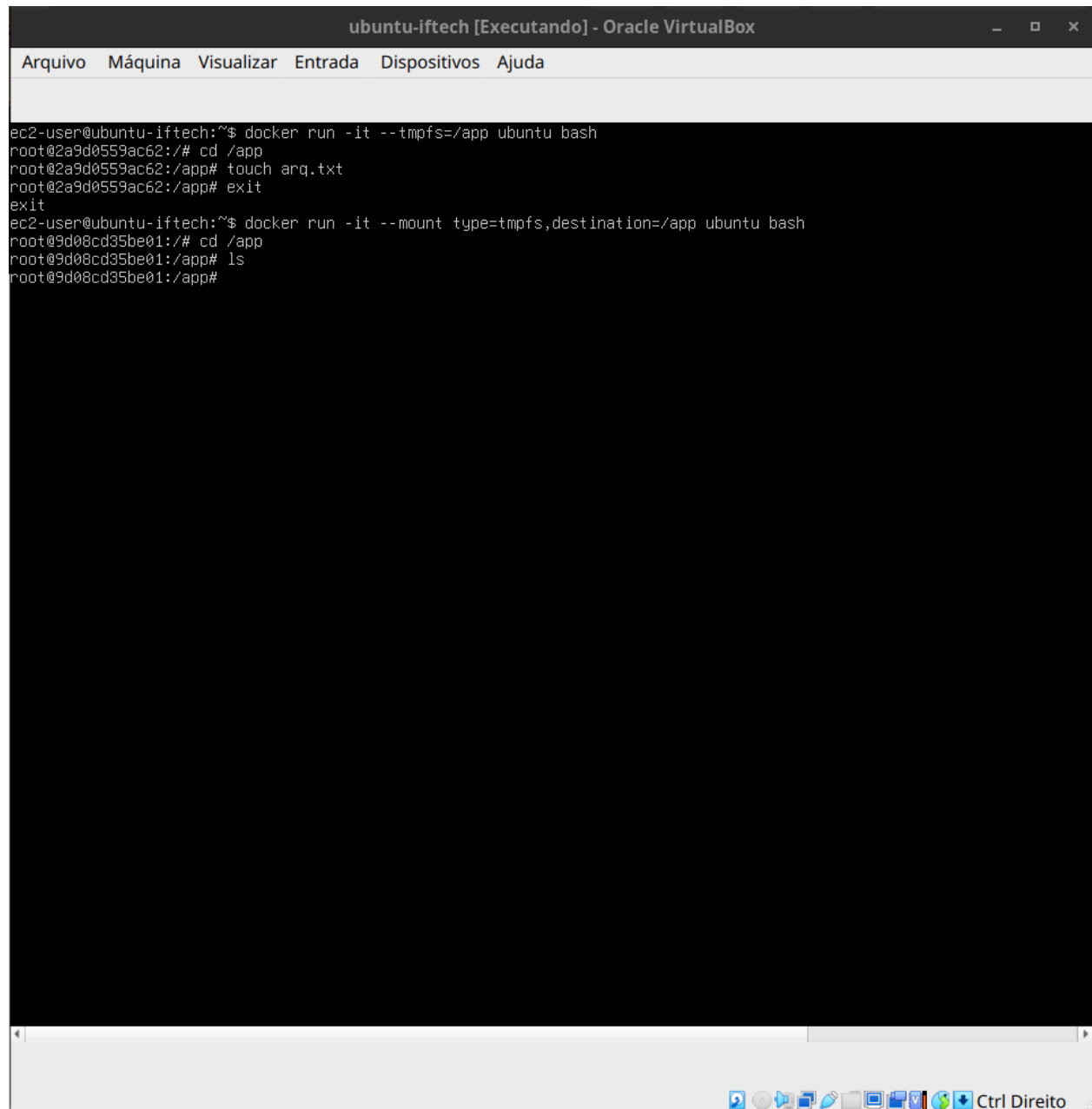
17) Criar o volume



```
ubuntu-iftech [Executando] - Oracle VirtualBox
Arquivo  Máquina  Visualizar  Entrada  Dispositivos  Ajuda

ec2-user@ubuntu-iftech:~$ docker volume create meu-volume
meu-volume
ec2-user@ubuntu-iftech:~$ docker volume ls
DRIVER      VOLUME NAME
local       app_ollama_data
local       meu-volume
ec2-user@ubuntu-iftech:~$ docker run -it -v meu-volume:/app ubuntu bash
root@25c2670ea1bb:/# cd /app
root@25c2670ea1bb:/app# touch um-art.txt
root@25c2670ea1bb:/app# exit
exit
ec2-user@ubuntu-iftech:~$ ls /var/lib/docker/volumes
ls: cannot access '/var/lib/docker/volumes': Permission denied
ec2-user@ubuntu-iftech:~$ sudo ls /var/lib/docker/volumes
[sudo] password for ec2-user:
app_ollama_data  backingFsBlockDev  metadata.db  meu-volume
ec2-user@ubuntu-iftech:~$ sudo ls /var/lib/docker/volumes/meu-volume/_data
um-art.txt
ec2-user@ubuntu-iftech:~$ _
```

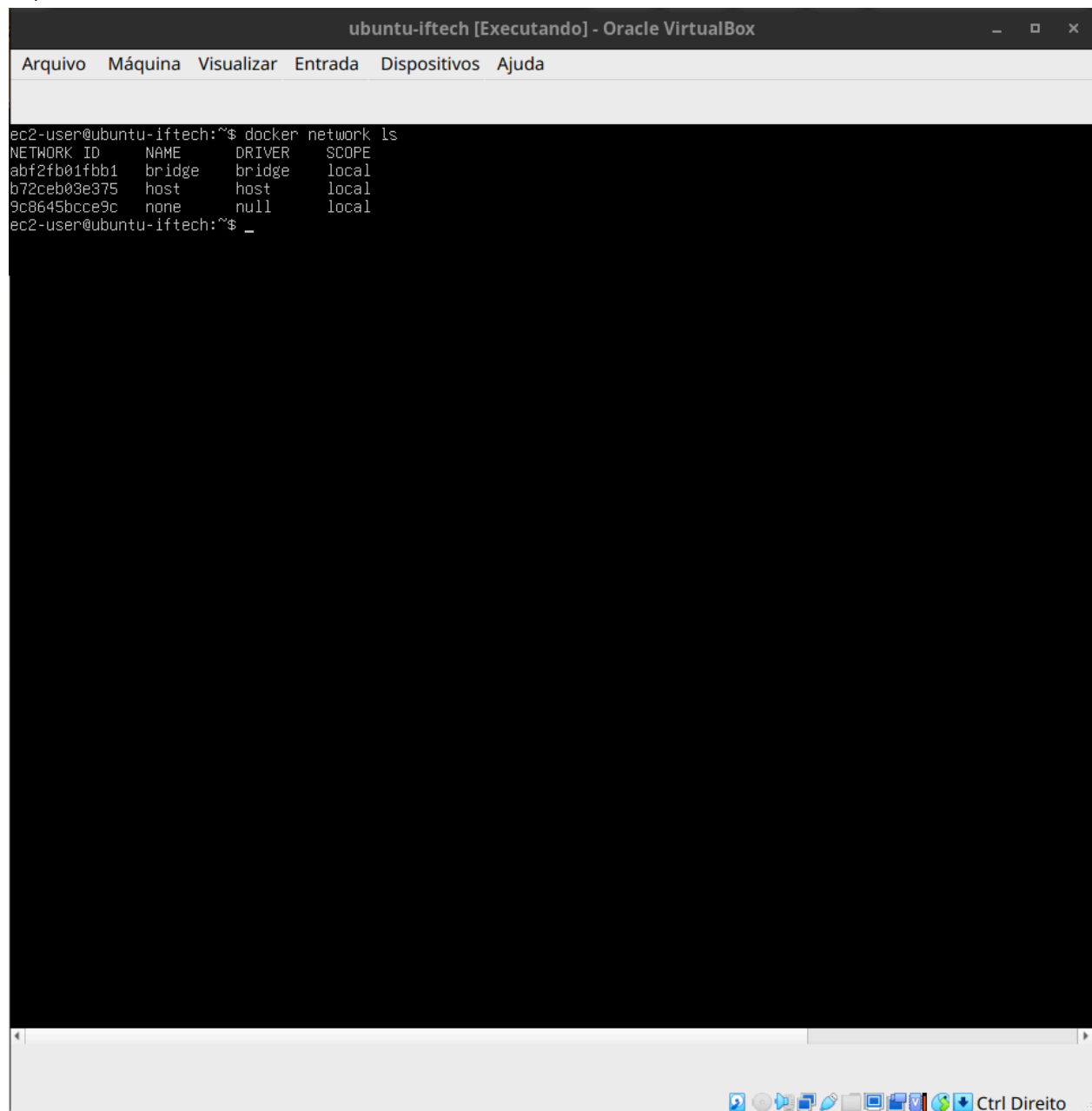
17.1) Criar o volume em memória



```
ubuntu-iftech [Executando] - Oracle VirtualBox
Arquivo  Máquina  Visualizar  Entrada  Dispositivos  Ajuda

ec2-user@ubuntu-iftech:~$ docker run -it --tmpfs=/app ubuntu bash
root@2a9d0559ac62:/# cd /app
root@2a9d0559ac62:/app# touch arq.txt
root@2a9d0559ac62:/app# exit
exit
ec2-user@ubuntu-iftech:~$ docker run -it --mount type=tmpfs,destination=/app ubuntu bash
root@9d08cd35be01:/# cd /app
root@9d08cd35be01:/app# ls
root@9d08cd35be01:/app#
```

18) Listar as redes do docker

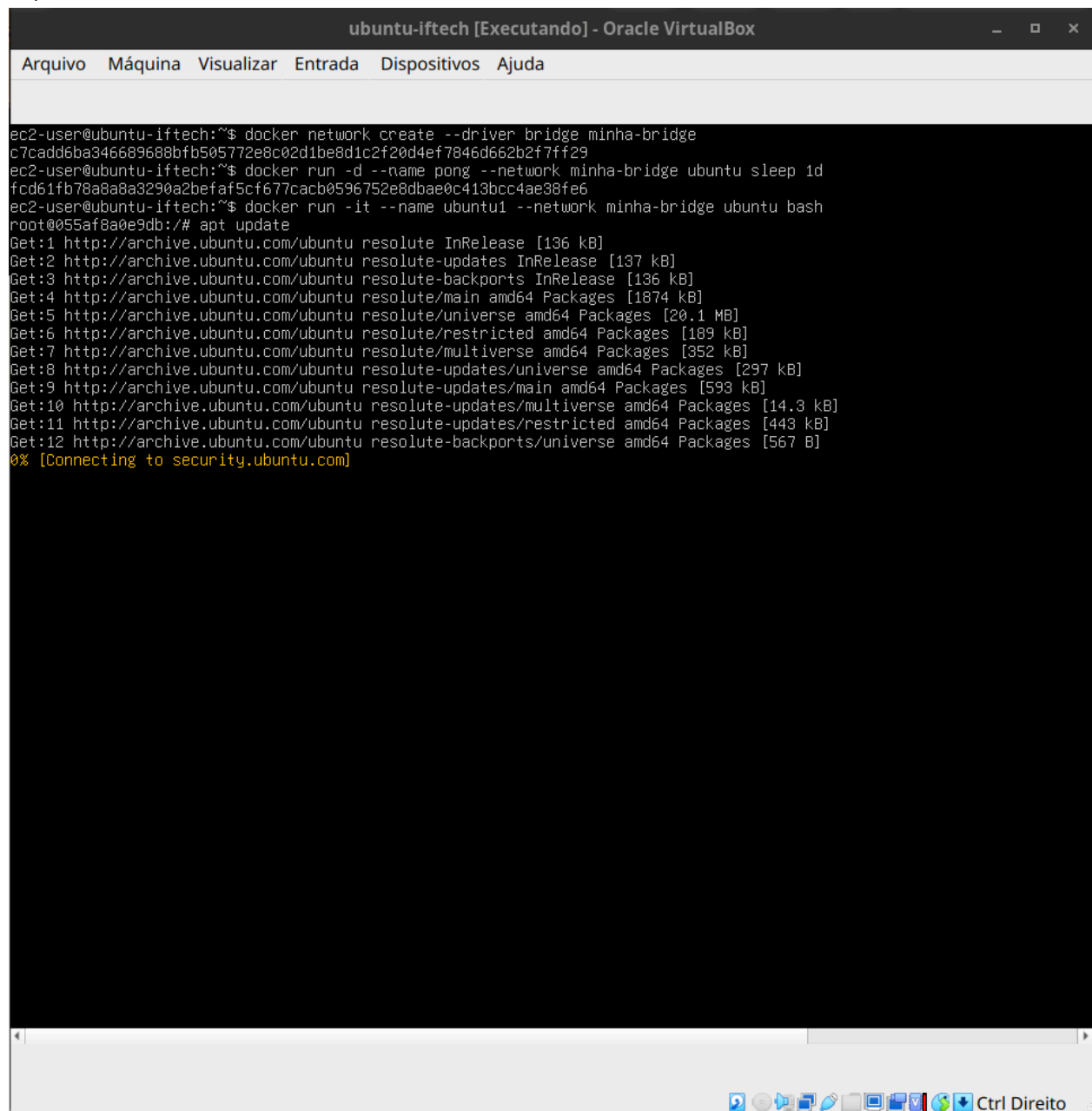


The screenshot shows a terminal window titled "ubuntu-iftech [Executando] - Oracle VirtualBox". The terminal displays the command `docker network ls` and its output, which lists three Docker networks. The output is as follows:

NETWORK ID	NAME	DRIVER	SCOPE
abf2fb01fbb1	bridge	bridge	local
b72ceb03e375	host	host	local
9c8645bce9c	none	null	local

The terminal prompt is `ec2-user@ubuntu-iftech:~$`. The window includes a menu bar with "Arquivo", "Máquina", "Visualizar", "Entrada", "Dispositivos", and "Ajuda". The bottom status bar shows various icons and the text "Ctrl Direito".

19) Criar a interface de rede



```
ubuntu-iftech [Executando] - Oracle VirtualBox
Arquivo  Máquina  Visualizar  Entrada  Dispositivos  Ajuda

ec2-user@ubuntu-iftech:~$ docker network create --driver bridge minha-bridge
c7cadd6ba346689688bfb505772e8c02d1be8d1c2f20d4ef7846d662b2f7ff29
ec2-user@ubuntu-iftech:~$ docker run -d --name pong --network minha-bridge ubuntu sleep 1d
fcd61fb78a8a8a3290a2befaf5cf677cacb0596752e8dbae0c413bcc4ae38fe6
ec2-user@ubuntu-iftech:~$ docker run -it --name ubuntu1 --network minha-bridge ubuntu bash
root@055af8a0e9db:/# apt update
Get:1 http://archive.ubuntu.com/ubuntu resolute InRelease [136 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [136 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute/main amd64 Packages [1874 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute/universe amd64 Packages [20.1 MB]
Get:6 http://archive.ubuntu.com/ubuntu resolute/restricted amd64 Packages [189 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute/multiverse amd64 Packages [352 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [297 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [593 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [14.3 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [443 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Packages [567 B]
0% [Connecting to security.ubuntu.com]
```

```
ubuntu-iftech [Executando] - Oracle VirtualBox
Arquivo  Máquina  Visualizar  Entrada  Dispositivos  Ajuda

root@055af8a0e9db:/# ping pong
PING pong (172.18.0.2) 56(84) bytes of data.
64 bytes from pong.minha-bridge (172.18.0.2): icmp_seq=1 ttl=64 time=0.089 ms
64 bytes from pong.minha-bridge (172.18.0.2): icmp_seq=2 ttl=64 time=0.079 ms
64 bytes from pong.minha-bridge (172.18.0.2): icmp_seq=3 ttl=64 time=0.108 ms
64 bytes from pong.minha-bridge (172.18.0.2): icmp_seq=4 ttl=64 time=0.060 ms
64 bytes from pong.minha-bridge (172.18.0.2): icmp_seq=5 ttl=64 time=0.088 ms
^C
--- pong ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4106ms
rtt min/avg/max/mdev = 0.060/0.084/0.108/0.015 ms
root@055af8a0e9db:/# docker ps
bash: docker: command not found
root@055af8a0e9db:/# exit
exit
ec2-user@ubuntu-iftech:~$ docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS
NAMES
fcd61fb78a8a   ubuntu    "sleep 1d"              About a minute ago    Up About a minute
pong
d2edab11495b   rcon/app-node:1.1   "npm start"            28 minutes ago       Up 28 minutes    0.0.0.0:905
upbeat_feistel
2ca44db5f8ea   rcon/app-node:1.0   "npm start"            35 minutes ago       Up 35 minutes    0.0.0.0:808
fervent_matsumoto
19deb58d6c8d   dockersamples/static-site  "/bin/sh -c 'cd /usr...'" About an hour ago     Up 53 minutes    443/tcp, 0.
pedantic_varahamihira
ec2-user@ubuntu-iftech:~$ _
```

20) Abrir a página web do container sem fazer o mapeamento de portas

```
ubuntu-iftech [Executando] - Oracle VirtualBox
Arquivo  Máquina  Visualizar  Entrada  Dispositivos  Ajuda

ec2-user@ubuntu-iftech:~$ docker run -d --network host dockersamples/static-site
242243a68d7c0b31e551e2f06d4b86897a502010d93a706d36b7393fb5343f8c
ec2-user@ubuntu-iftech:~$ wget http://localhost:80
--2026-08-18 12:12:38--  http://localhost/
Resolving localhost (localhost)... ::1, 127.0.0.1
Connecting to localhost (localhost)|::1|:80... failed: Connection refused.
Connecting to localhost (localhost)|127.0.0.1|:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 8749 (8,5K) [text/html]
Saving to: 'index.html.1'

index.html.1                                100%[=====]

2026-08-18 12:12:38 (272 MB/s) - 'index.html.1' saved [8749/8749]

ec2-user@ubuntu-iftech:~$ head -n 15 index.html.1
<!DOCTYPE html><html>

<head>
<meta charset="utf-8">
<style type="text/css">
body {
  font-family: Helvetica, arial, sans-serif;
  font-size: 14px;
  line-height: 1.6;
  padding-top: 10px;
  padding-bottom: 10px;
  background-color: white;
  padding: 30px; }

body > *:first-child {
ec2-user@ubuntu-iftech:~$ tail -n 15 index.html.1
<br>
<br>
<br>
<br>
<br></p>

<h1 id="toc_0">Hello Docker!</h1>

<p>This is being served from a <b>docker</b><br>
container running Nginx.</p>

</body>

</html>
ec2-user@ubuntu-iftech:~$ _
```

21) Instalação do docker-compose

The screenshot shows a Google search result for the query "o docker-compose com hífen ficou obsoleto". The search bar at the top contains the query and navigation icons. Below the search bar, the "All" tab is selected. The main content area features an "AI Overview" section with a blue star icon. The text in this section states that the command `docker-compose` (with a hyphen) has been replaced by **Docker Compose V2**, which uses `docker compose` (without a hyphen) integrated into the Docker CLI. It also mentions that the old version in Python stopped receiving updates in 2023. Below this, a section titled "Mudanças importantes" (Important Changes) lists three bullet points: 1. "Comando novo: Use `docker compose` sem hífen." (New command: Use `docker compose` without a hyphen.) 2. "Fim do suporte V1: A versão antiga foi descontinuada." (End of V1 support: The old version has been discontinued.) 3. "Nome do arquivo: O arquivo ainda pode se chamar `docker-compose.yml`, mas a chave `version:` dentro dele virou obsoleta." (File name: The file can still be named `docker-compose.yml`, but the `version:` key inside it has become obsolete.) Below the list, a summary sentence says "Um resumo rápido e prático de como o comando atual funciona na prática:" (A quick and practical summary of how the current command works in practice:). This is followed by a video player for a video titled "Descomplicando o docker compose" by Rocketseat, which has 3.6K views and is 8 months old. At the bottom, a section titled "Se você quiser, posso te ajudar a:" (If you want, I can help you:) lists two more bullet points: 1. "Atualizar scripts antigos que ainda usam o hífen" (Update old scripts that still use the hyphen) and 2. "Remover a linha de versão obsoleta dos seus arquivos `.yml`" (Remove the obsolete version line from your `.yml` files). The bottom of the image shows a Windows taskbar with various application icons.

o docker-compose com hífen ficou obsoleto

AI Overview

Sim, o comando `docker-compose` (com hífen) foi substituído pelo **Docker Compose V2**, que usa `docker compose` (sem hífen) integrado direto no CLI do Docker. A versão antiga em Python parou de receber atualizações em 2023. Reddit +3

Mudanças importantes

- **Comando novo:** Use `docker compose` sem hífen.
- **Fim do suporte V1:** A versão antiga foi descontinuada.
- **Nome do arquivo:** O arquivo ainda pode se chamar `docker-compose.yml`, mas a chave `version:` dentro dele virou obsoleta. Alura +3

Um resumo rápido e prático de como o comando atual funciona na prática:

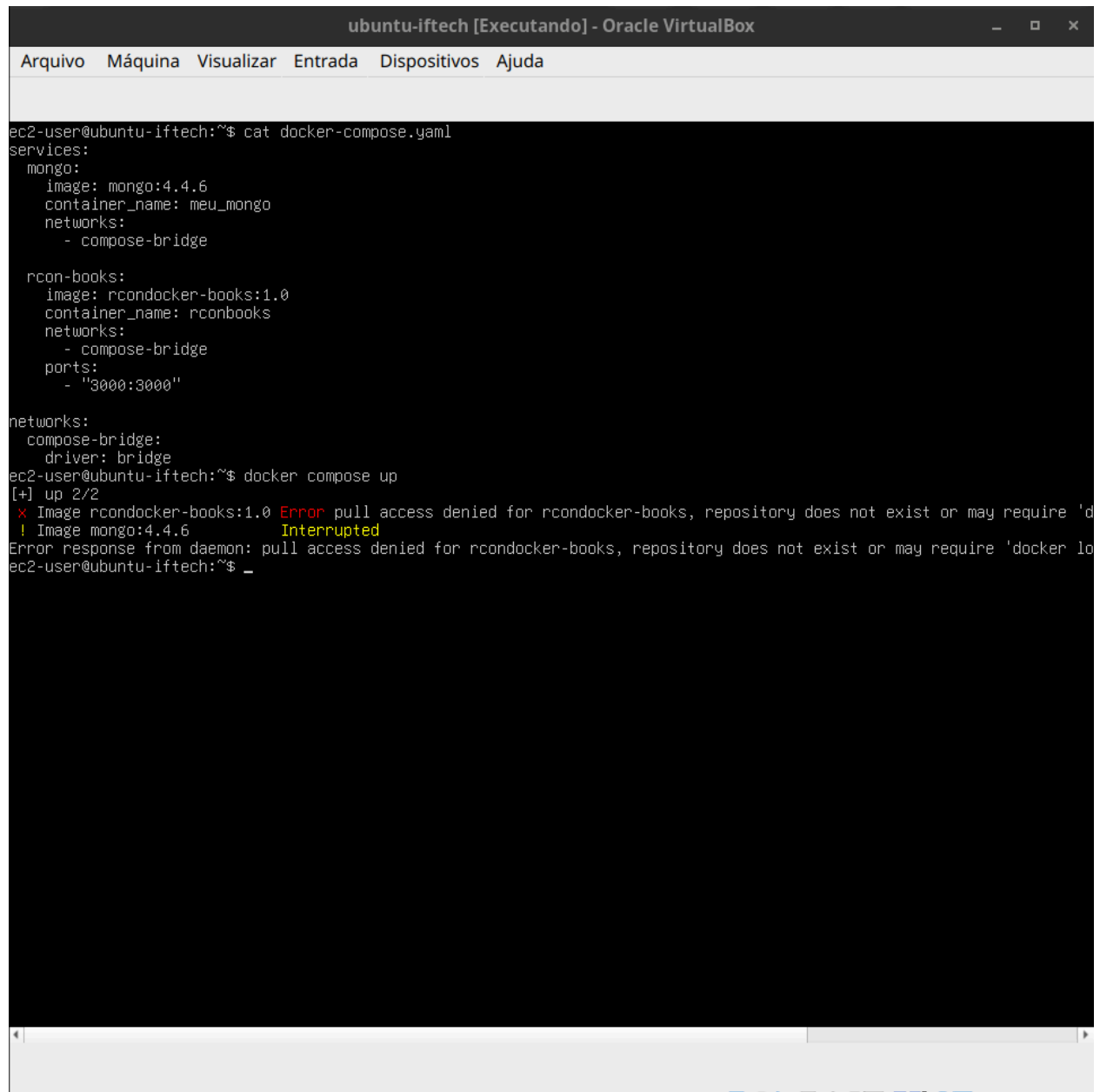
Descomplicando o docker compose

3.6K views · 8 months ago
YouTube · Rocketseat

Se você quiser, posso te ajudar a:

- **Atualizar scripts** antigos que ainda usam o hífen
- **Remover a linha de versão** obsoleta dos seus arquivos `.yml`

22) Imagem não está com permissão de pull



```
ubuntu-iftech [Executando] - Oracle VirtualBox
Arquivo  Máquina  Visualizar  Entrada  Dispositivos  Ajuda

ec2-user@ubuntu-iftech:~$ cat docker-compose.yaml
services:
  mongo:
    image: mongo:4.4.6
    container_name: meu_mongo
    networks:
      - compose-bridge

  rcon-books:
    image: rcondocker-books:1.0
    container_name: rconbooks
    networks:
      - compose-bridge
    ports:
      - "3000:3000"

networks:
  compose-bridge:
    driver: bridge
ec2-user@ubuntu-iftech:~$ docker compose up
[+] up 2/2
x Image rcondocker-books:1.0 Error pull access denied for rcondocker-books, repository does not exist or may require 'd
! Image mongo:4.4.6 Interrupted
Error response from daemon: pull access denied for rcondocker-books, repository does not exist or may require 'docker lo
ec2-user@ubuntu-iftech:~$ _
```